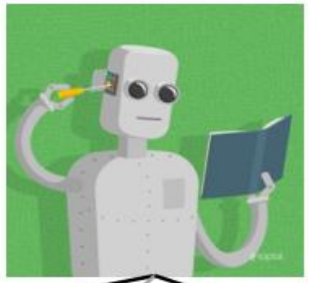
This week

- HW4, Lab 4 and Project 2 Data description due tonight
- Today: Clustering and review
- Today: Midterm assigned after class
  - Due by 11:59pm Thursday 10/30
  - Take-home
  - Don’t procrastinate

Clustering

Taxonomy of Machine Learning



Labeled Data

Reward

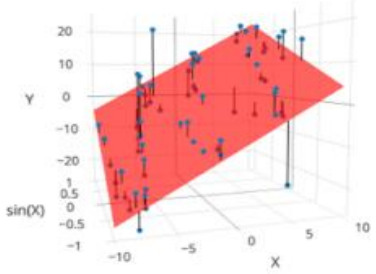
Unlabeled Data

Supervised Learning

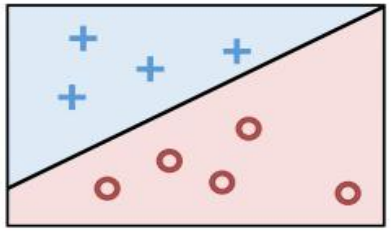
Quantitative Response

Categorical Response

Regression



Classification



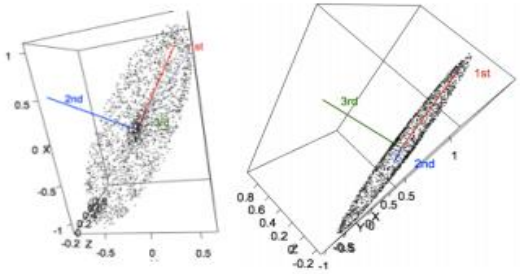
Reinforcement Learning

(not covered)

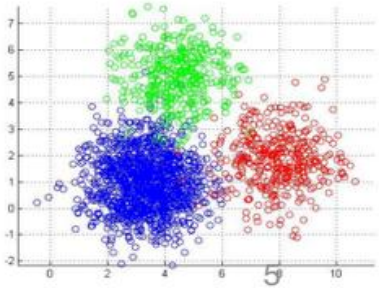
Alpha Go

Unsupervised Learning

Dimensionality Reduction



Clustering



Unsupervised Learning

With **Supervised Learning**, we have some sort of clear objective in mind with respect to a target variable.

- What's the strength of association between *y* and *x* predictors?
- How well can we predict y?

→ **Our modeling process is guided by these objectives**

Unsupervised Learning

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- What's the strength of association between *y* and *x* predictors?
- How well can we predict y?

  → **Our modeling process is guided by these objectives**

Unsupervised Learning has no such objectives

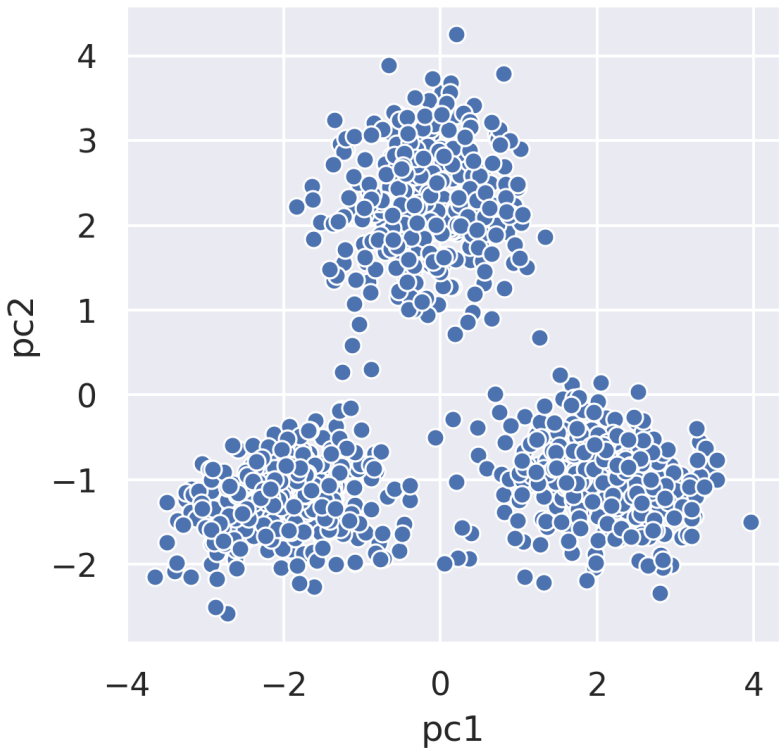
Unsupervised Learning

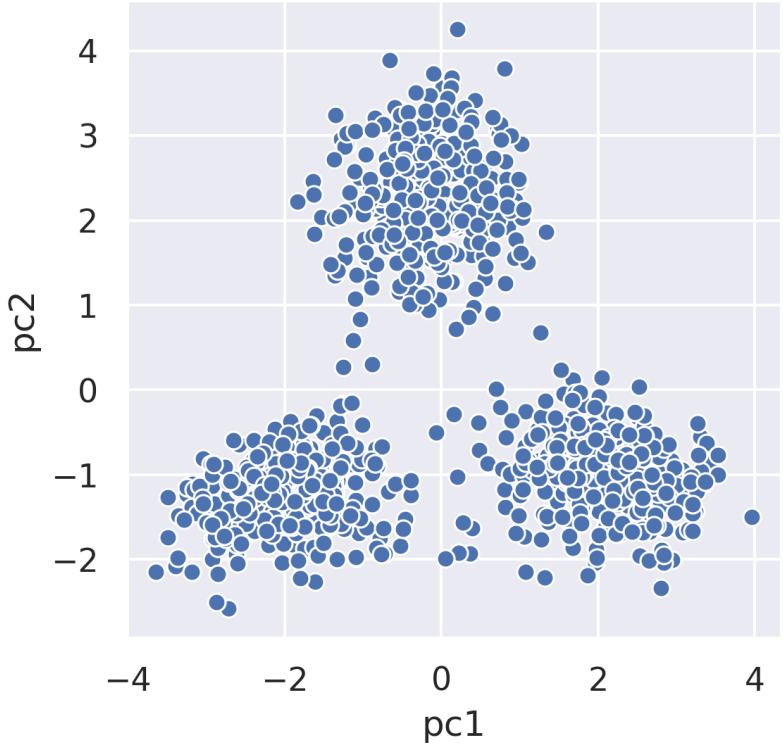
Unsupervised Learning is the process of revealing structure and information in the data w/o respect to any clear objective

- There is typically no target (*y* variable) involved.
  - Therefore no straightforward metrics of success
- Aiming for organic description of the data, often with respect to associations.
  - Categorical associations → Clustering
  - Quantitative associations → Ordination (e.g. Principal Components Analytics)

Clustering

Clustering can be thought of as a classification approach to unsupervised learning.

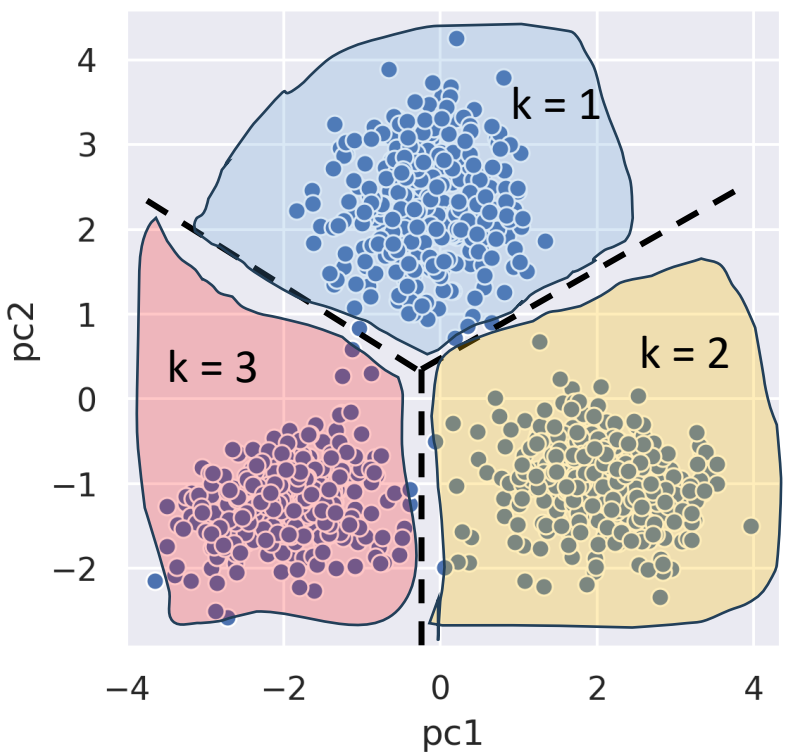


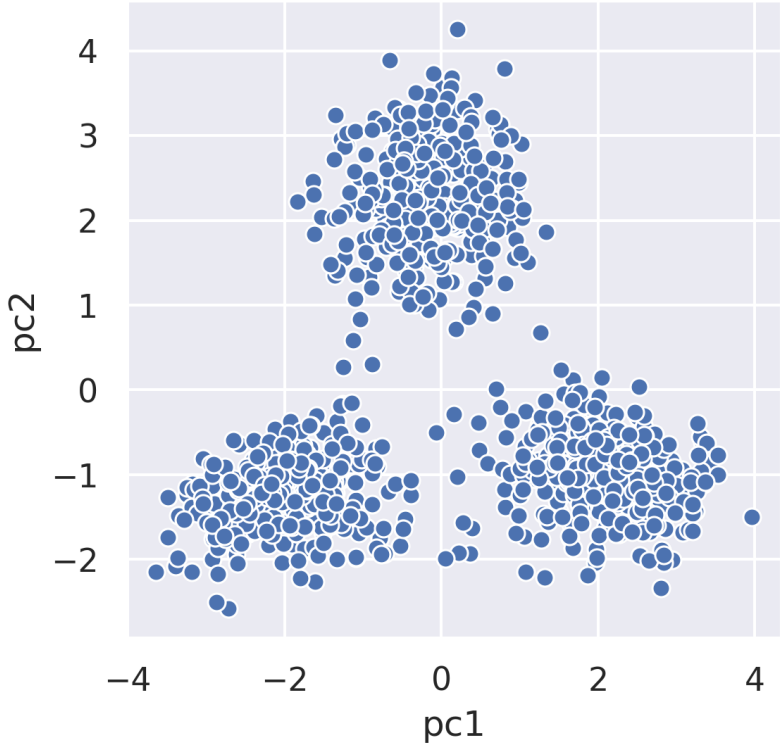


Clustering

Clustering can be thought of as a classification approach to unsupervised learning.

Classification (Supervised Learning)





Clustering

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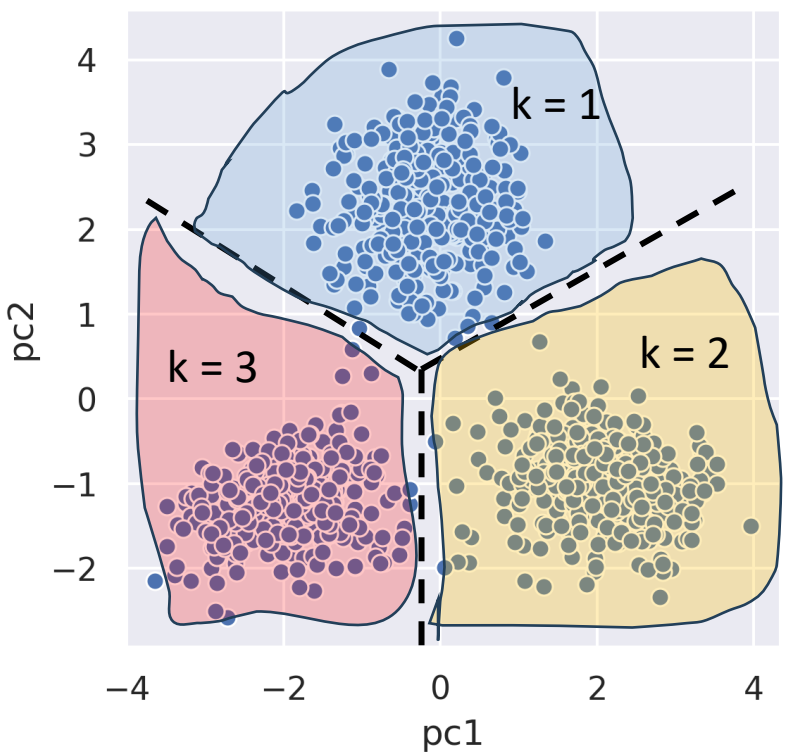
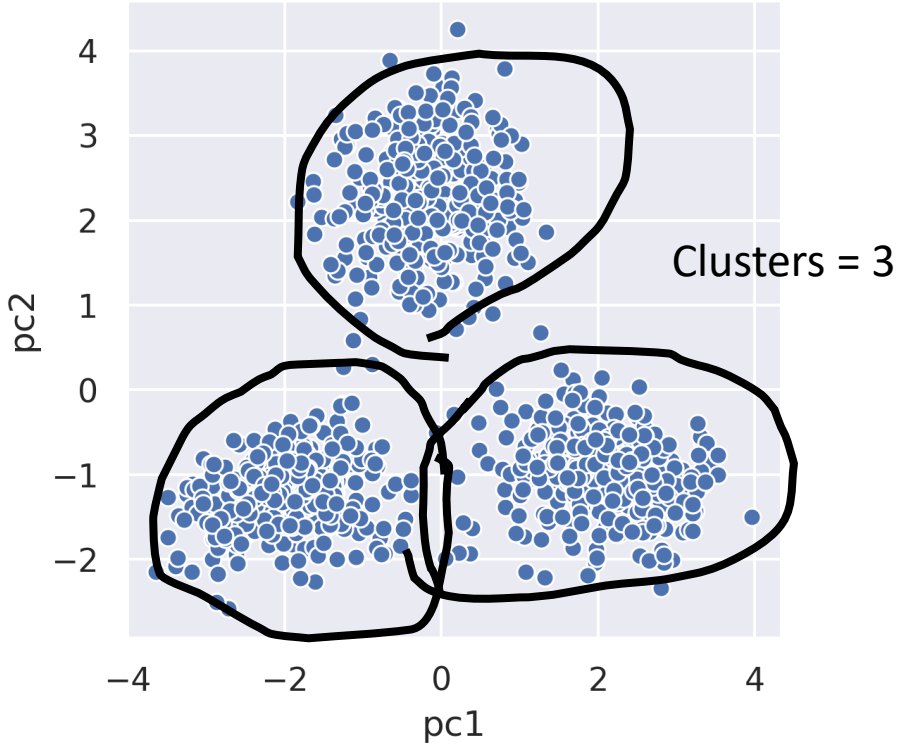
Classification (Supervised Learning)



Clustering (Unsupervised Learning)



Clustering Example

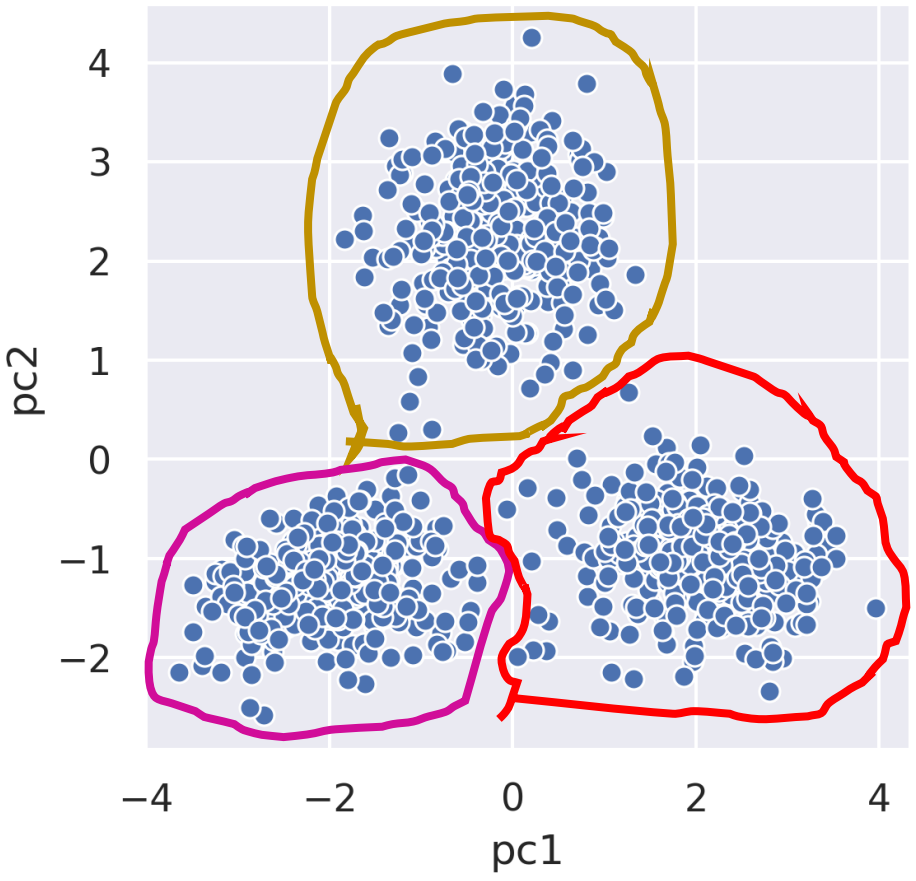
Consider the following figure

- Each point represents two dimensions of variable space on how much time patrons spent at 8 different zoo exhibits

Goal of clustering: assign each point to a cluster

This is an unsupervised task

- We don’t have labels for each visitor
- We want to infer pattern even without labels



Clustering Example: Netflix

Suppose you're a data scientist for Netflix and have information on customer viewing habits

- Can use clustering to assign each person or show to a "cluster"
  - Clusters could be types of viewers
  - Clusters could be types of shows
- Don't have to define clusters in advance

In supervised classification, we have predefined classes

- Clustering helps us "discover" classes instead

Clustering Example: Netflix

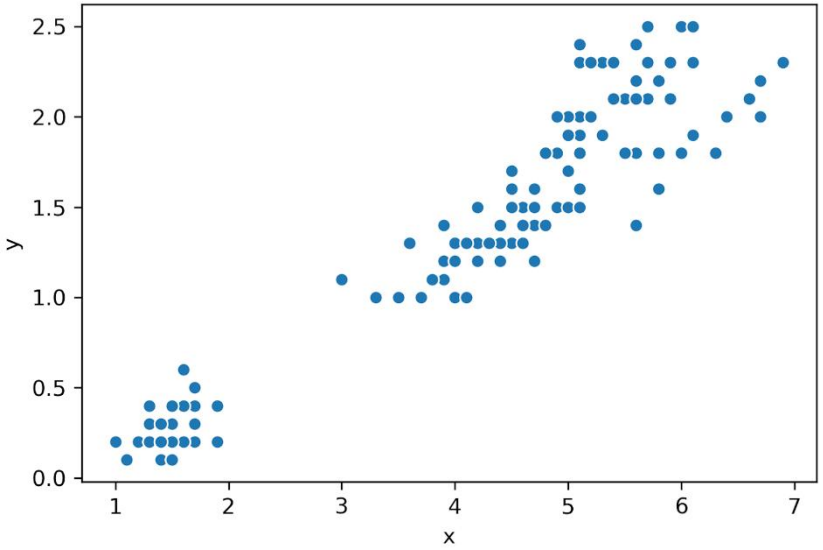
K-Means Clustering Algorithm

K-Means Clustering

Most popular clustering approach: K-Means

- Pick an arbitrary k, and **randomly place k "centers",** each a different color
- Repeat until convergence:
  - Color points according to the closest center
  - Move center for each color to center of points with that color

Raw data →



K-Means Clustering

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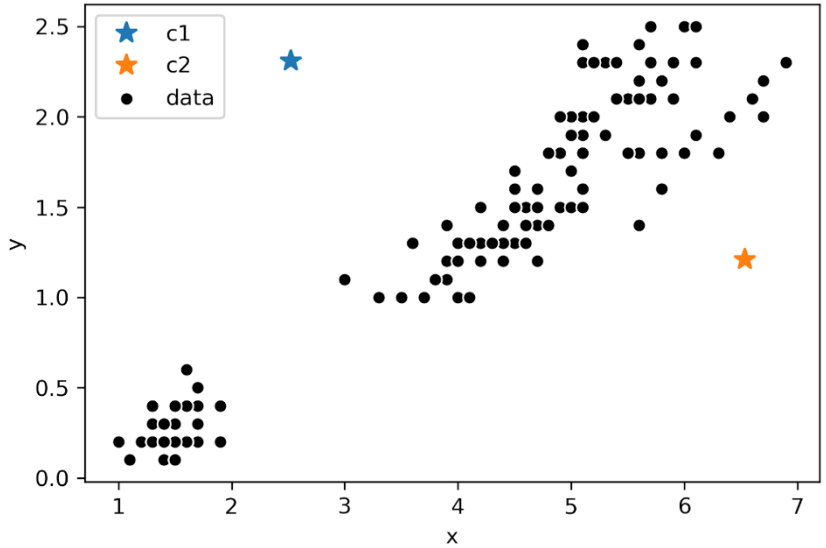
Initial random placement of two centers



Iteration: 0

K-Means Clustering

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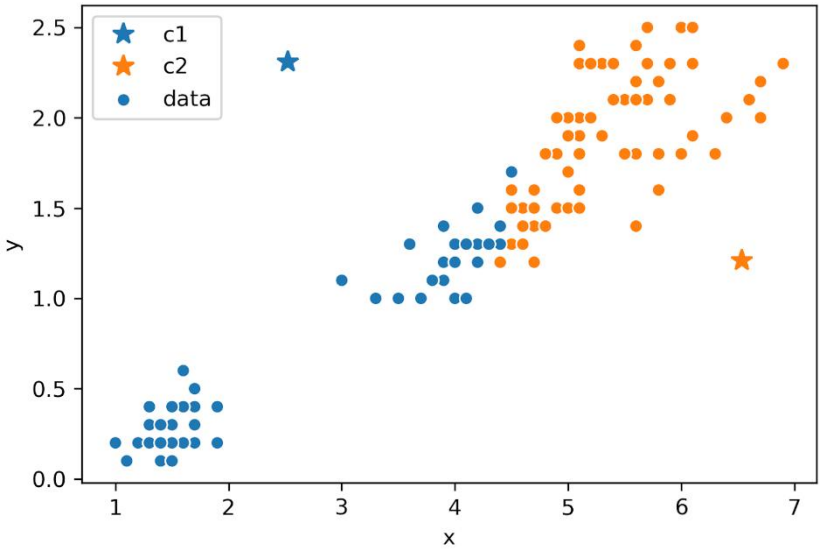
Data colored by closest center



Iteration: 0

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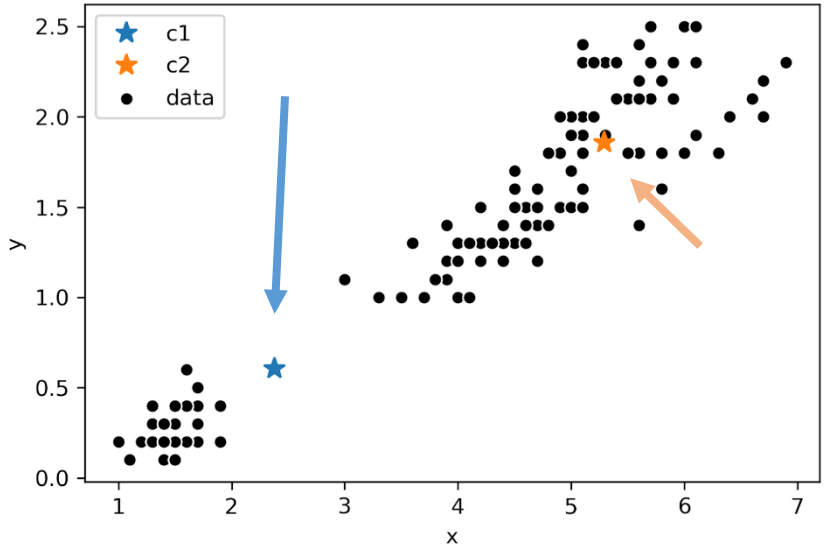
Centers move to their new homes



Iteration: 1

K-Means Clustering

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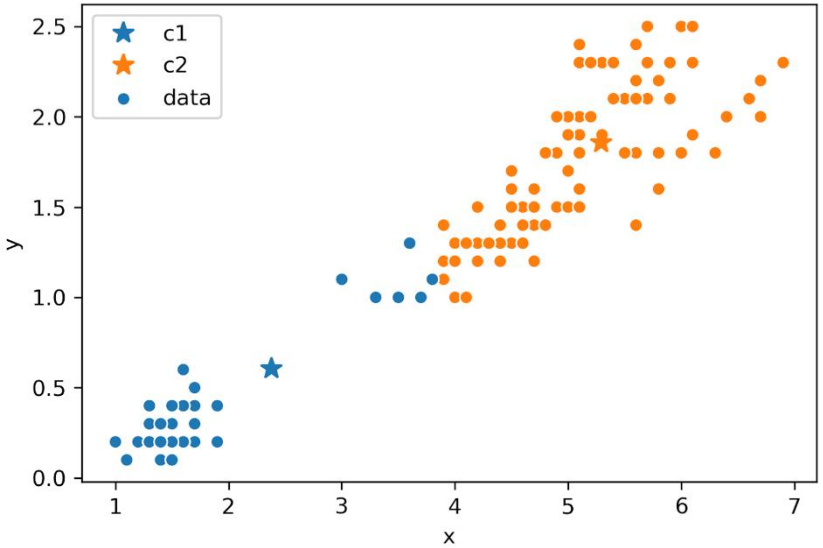
Data colored by closest center (in new position)



Iteration: 1

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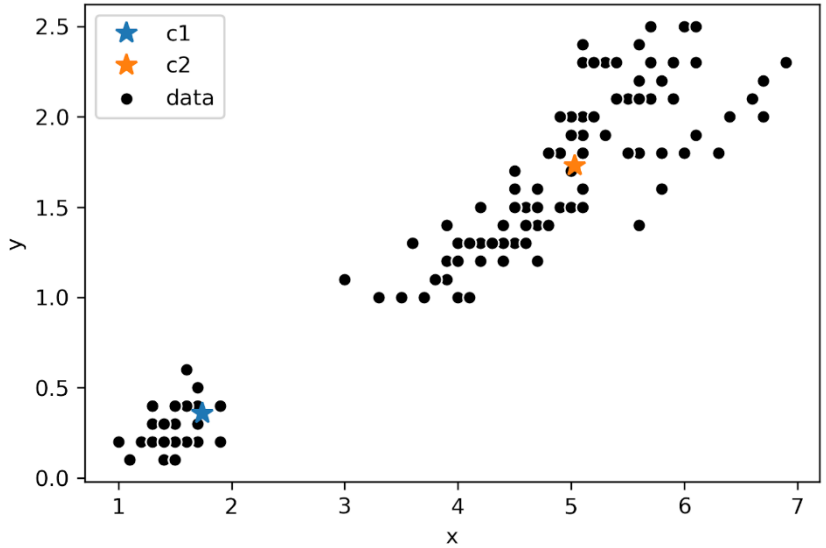
Centers moved to new position



Iteration: 2

K-Means Clustering

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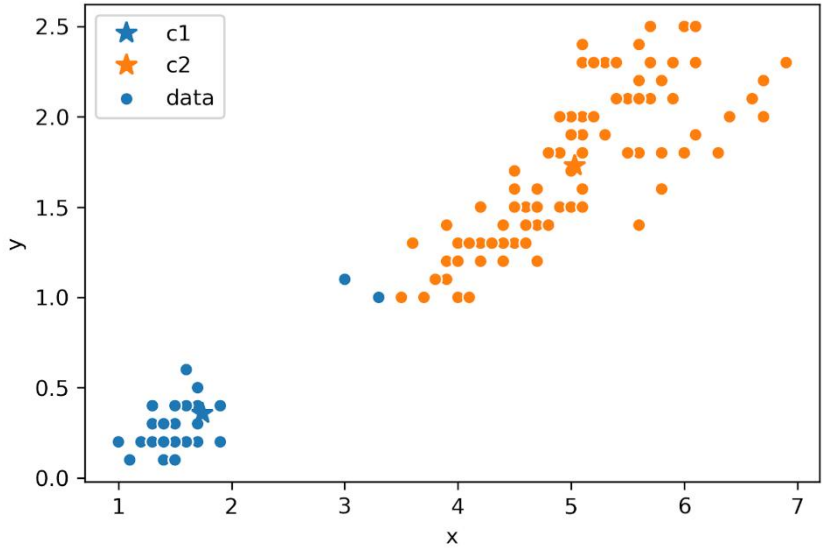
Data colored by closest center (in new position)



Iteration: 2

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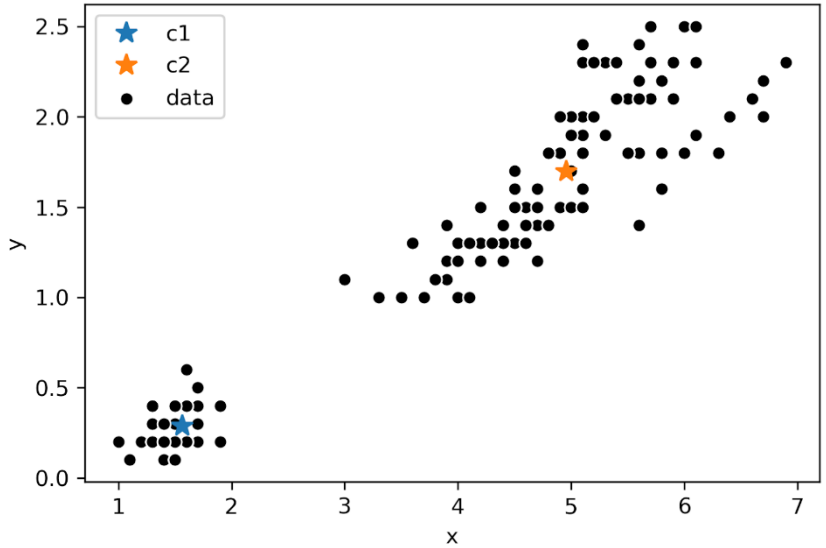

Iteration: 3

K-Means Clustering

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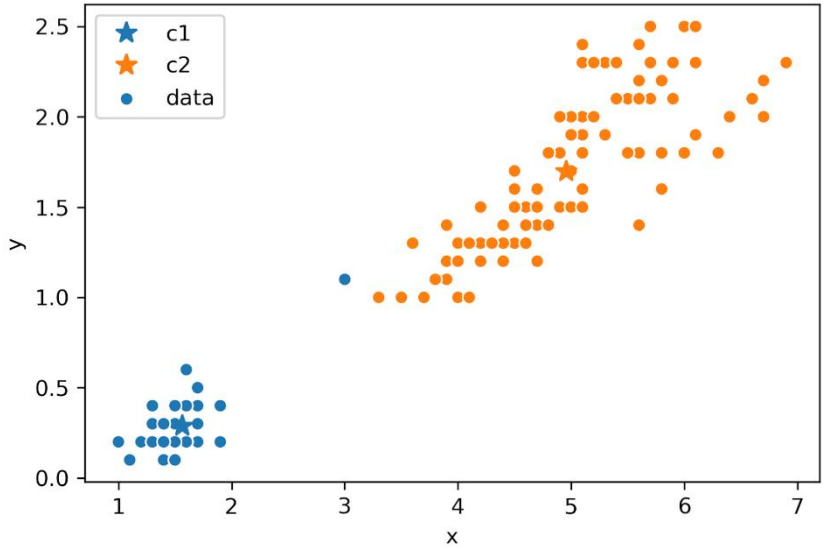
Data colored by closest center (in new position)



Iteration: 3

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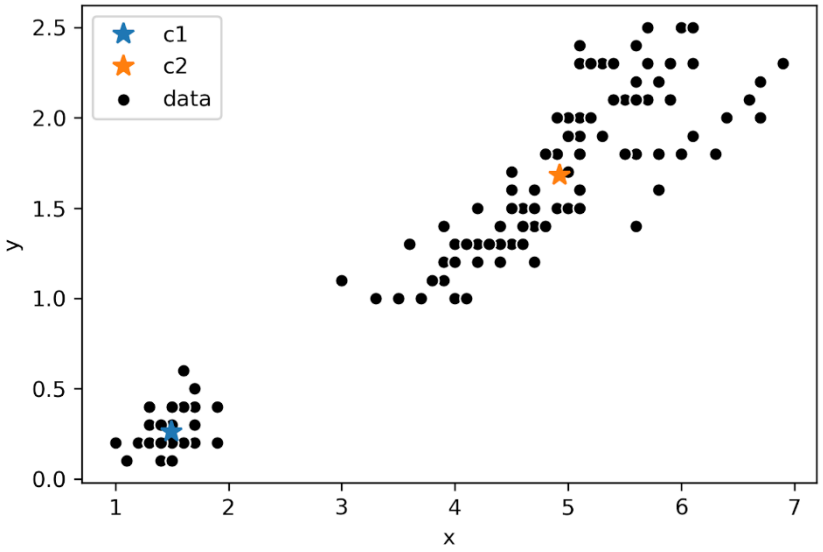

Iteration: 4

K-Means Clustering

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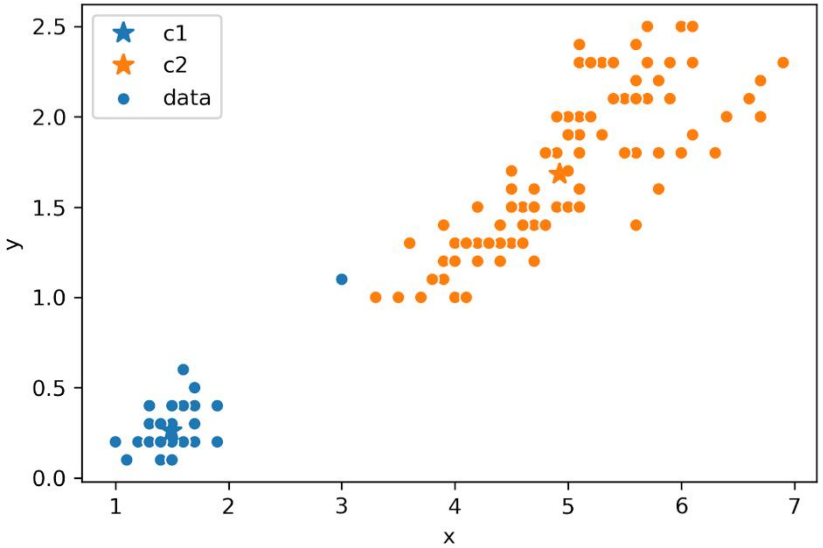
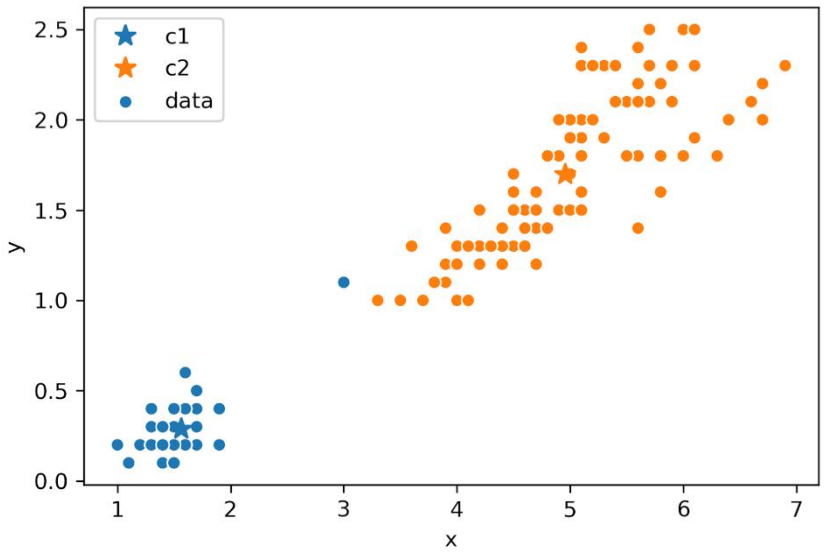
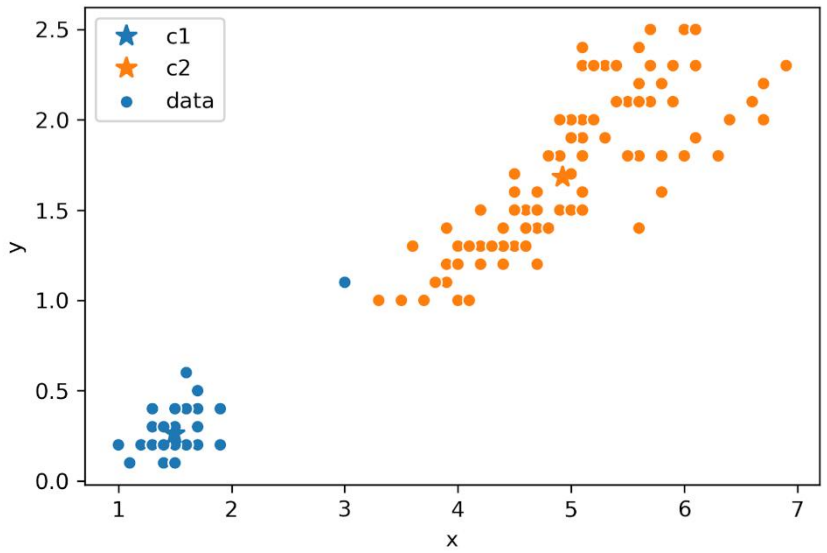


Iteration: 4

K-Means Clustering



Iteration: 4

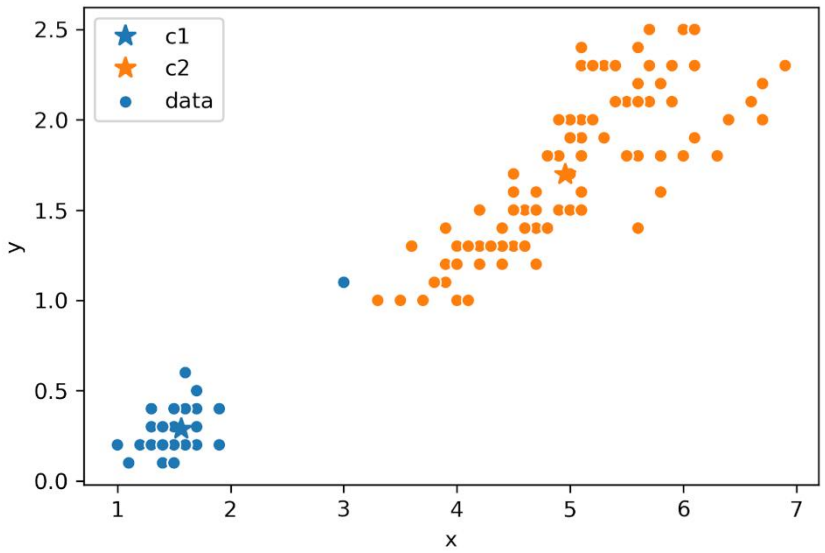


Iteration: 5

Above we see the results after iterations 4 and 5

- Centers moved slightly between iterations 4 and 5
- But no points changed color

K-Means Clustering



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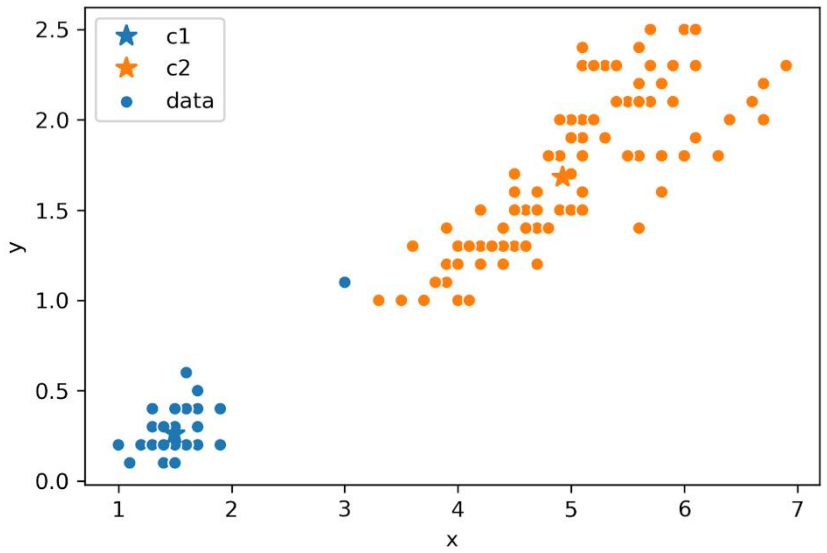

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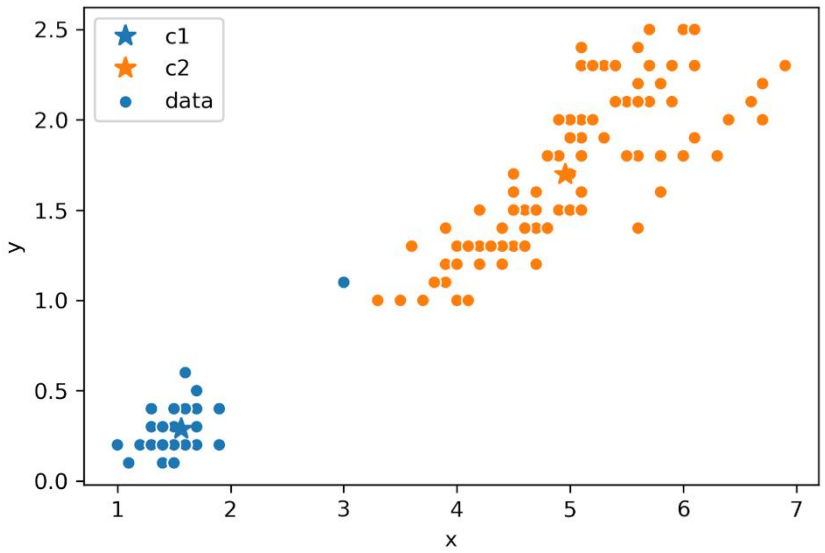
- Are we done?

K-Means Clustering



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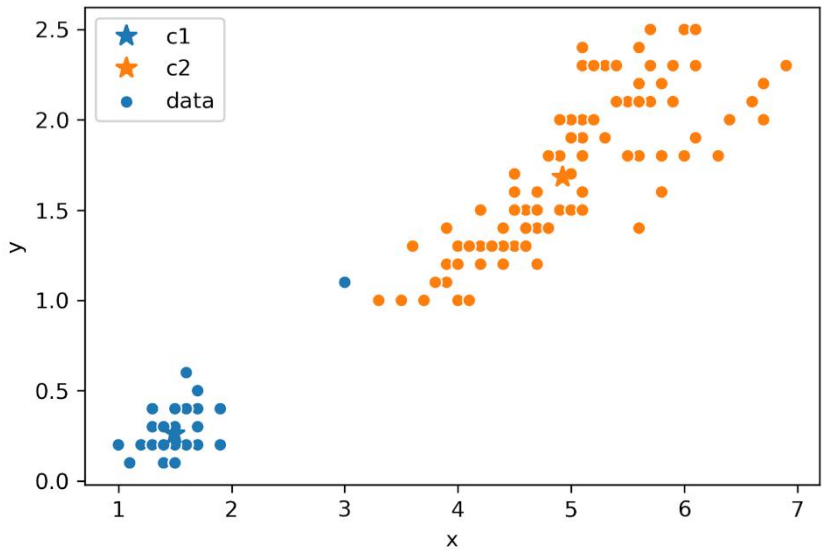

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Above we see the results after iterations 4 and 5

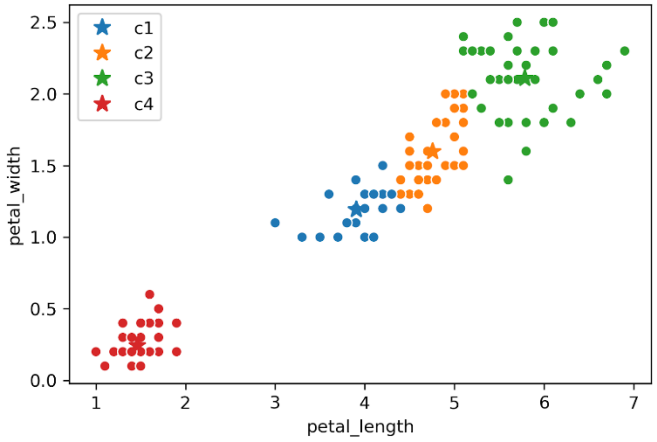
- Centers moved slightly between iterations 4 and 5
- But no points changed color
- Are we done?
  - Yes! If we tried iteration 6, we'd see that the centers don't move at all.

Minimizing Inertia

K-Means Clustering for K = 4

- Each time you run K-Means, you get a different output

random.seed(25)



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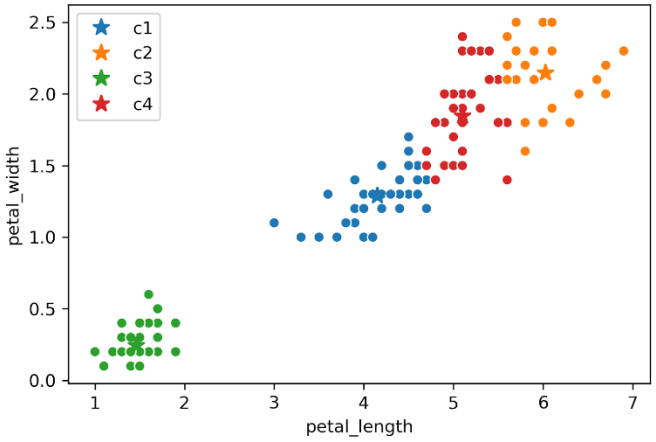
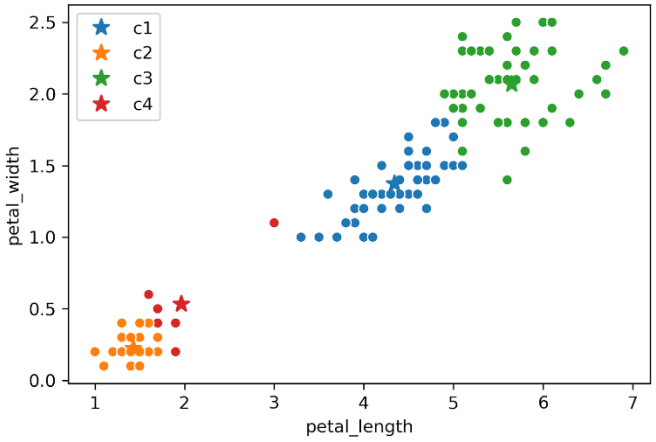


random.seed(40)



- Which is best?

K-Means Clustering for K = 4

- Each time you run K-Means, you get a different output

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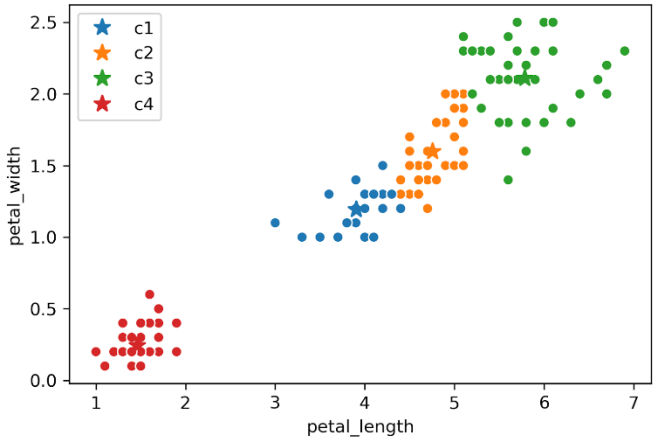

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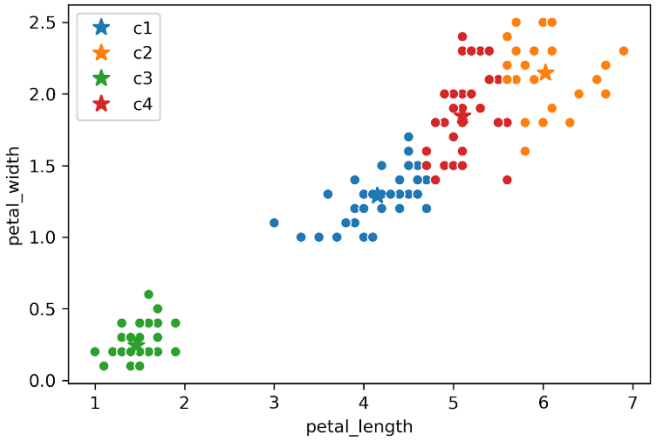

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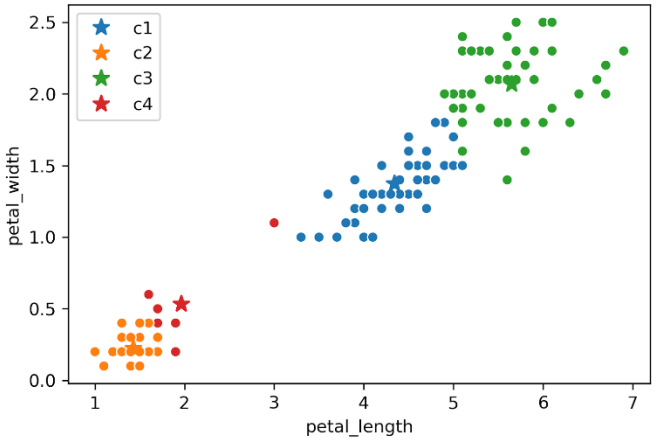


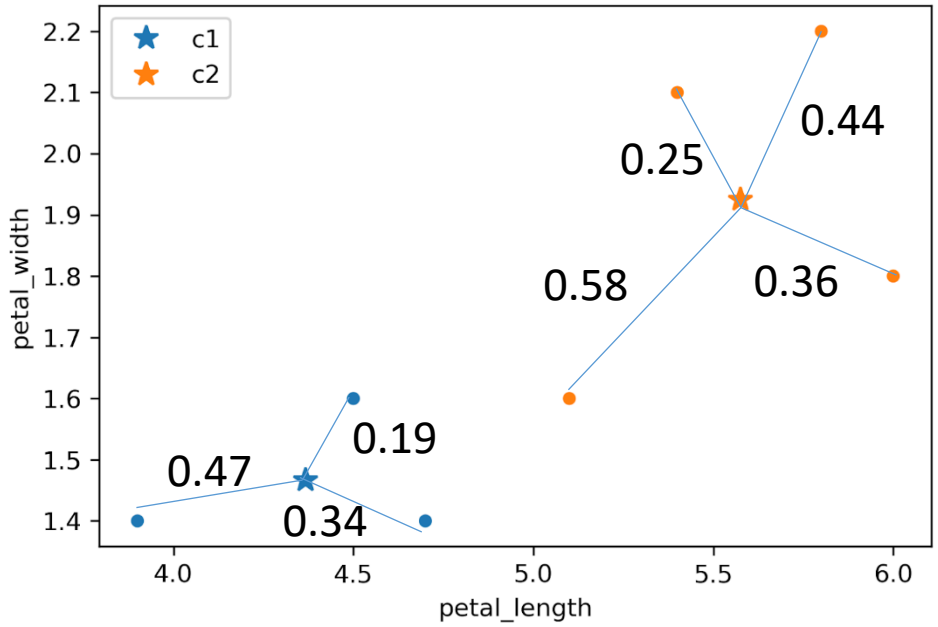
- Which is best?

- One approach: Define some sort of loss function

K-Means Clustering for K = 4

To evaluate different clustering results, we need a loss function

Two common loss functions:

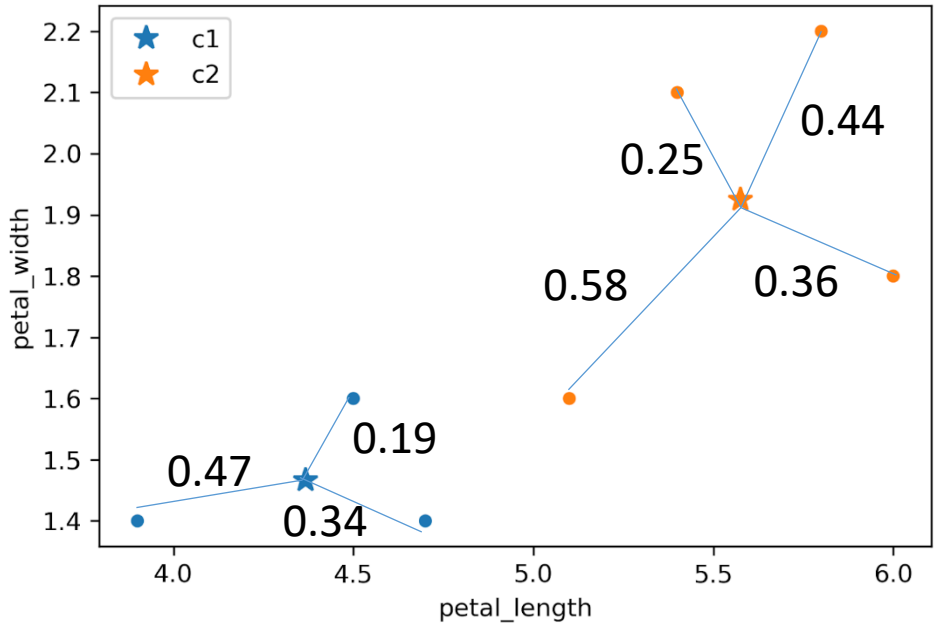


K-Means Clustering for K = 4

To evaluate different clustering results, we need a loss function

Two common loss functions:

- **Inertia**: Sum of squared distances from each data point to its center



Example:

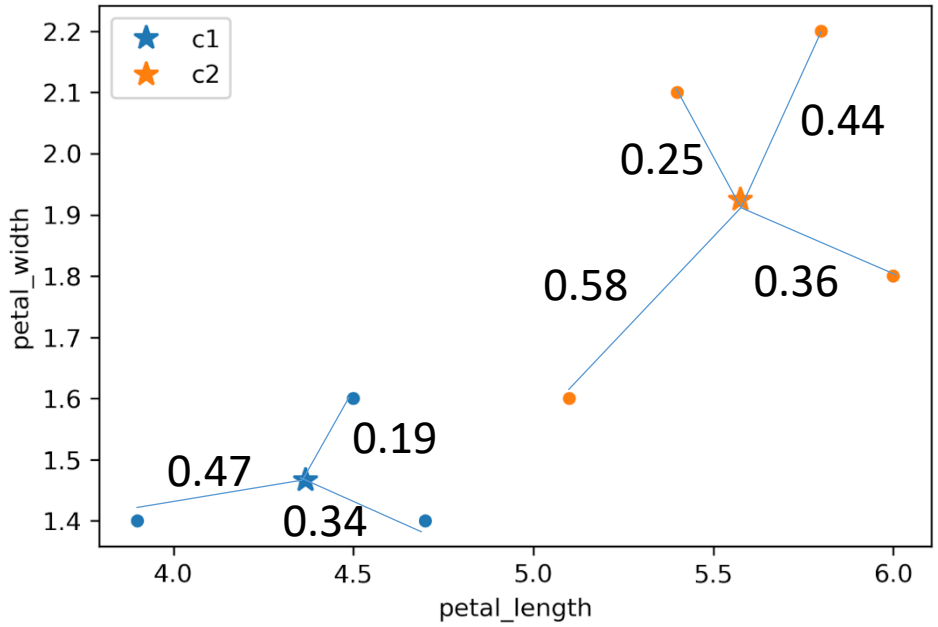
- **inertia: $0.47^2 + 0.19^2 + 0.34^2 + 0.25^2 + 0.58^2 + 0.36^2 + 0.44^2$**

K-Means Clustering for K = 4

To evaluate different clustering results, we need a loss function

Two common loss functions:

- Inertia: Sum of squared distances from each data point to its center
- **Distortion**: Weighted sum of squared distances from each data point to its center



Example:

- inertia: $0.47^2 + 0.19^2 + 0.34^2 + 0.25^2 + 0.58^2 + 0.36^2 + 0.44^2$
- **distortion:** $(0.47^2 + 0.19^2 + 0.34^2)/ 3 + (0.25^2 + 0.58^2 + 0.36^2 + 0.44^2) / 4$

K-Means Clustering for K = 4

- Each time you run K-Means, you get a different output

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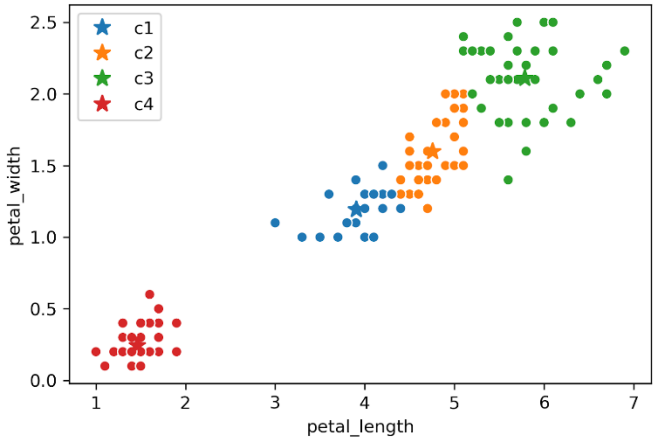

Inertia: 44.96

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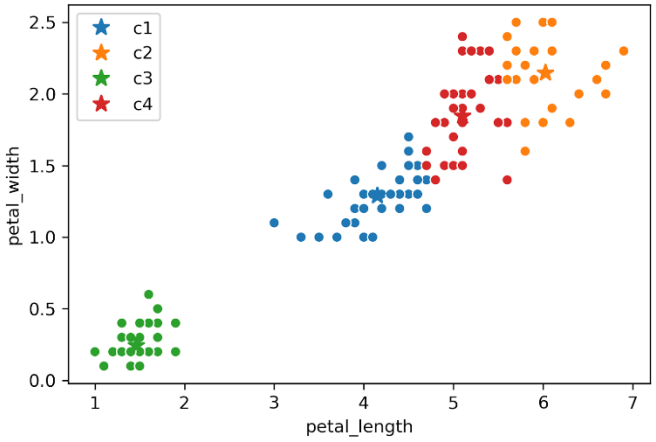
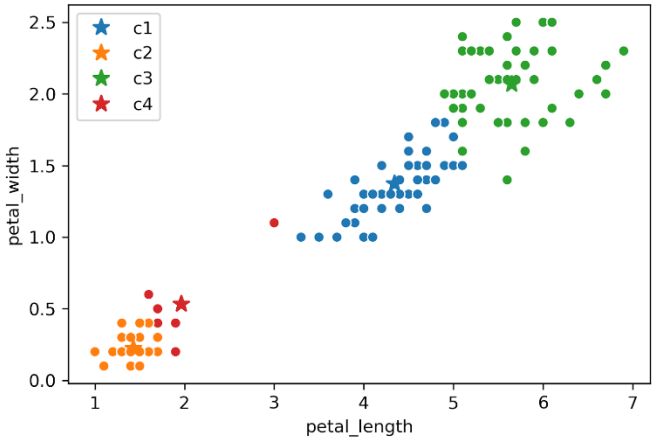


Inertia: 45.95

random.seed(40)



Inertia: 54.35

- Our loss function says that the leftmost clustering is best (inertia: 44.96) and rightmost clustering (inertia: 54.35) is worst.

K-Means and Inertia

The loss function K-Means is trying to minimize inertia

... but often fails to find a global optimum.

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Can think of K-Means as a pair of optimizers that take turns

- First optimizer:
  - Holds center positions constant
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K-Means and Inertia

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Can think of K-Means as a pair of optimizers that take turns

- First optimizer:
  - Holds center positions constant
  - Optimizes data colors
- Second optimizer:
  - Holds data colors constant
  - Optimizes center positions
- Neither optimizer has total control

Optimizing Inertia

Hard problem: Give an algorithm that optimizes inertia

- Your algorithm should return the EXACT best centers and colors
- Don’t worry about runtime

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Algorithm:

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  - Compute the inertia for the k centers
    - If current inertia is better than best known, write down the current centers and coloring and call that the new best known

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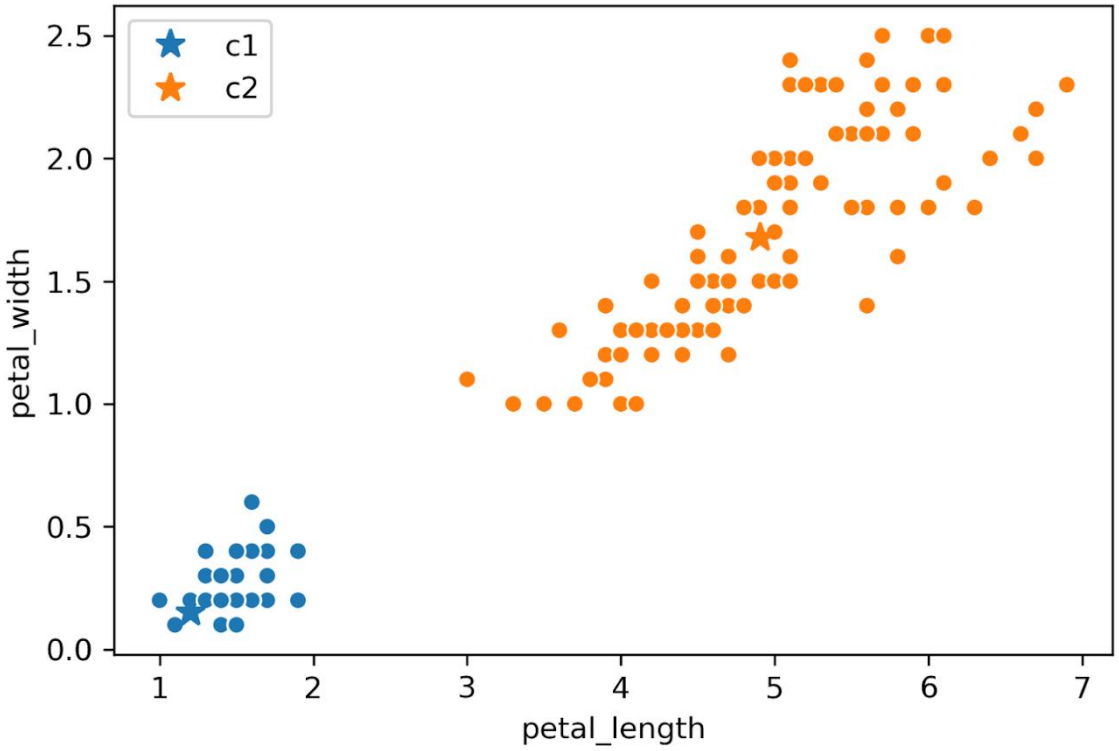
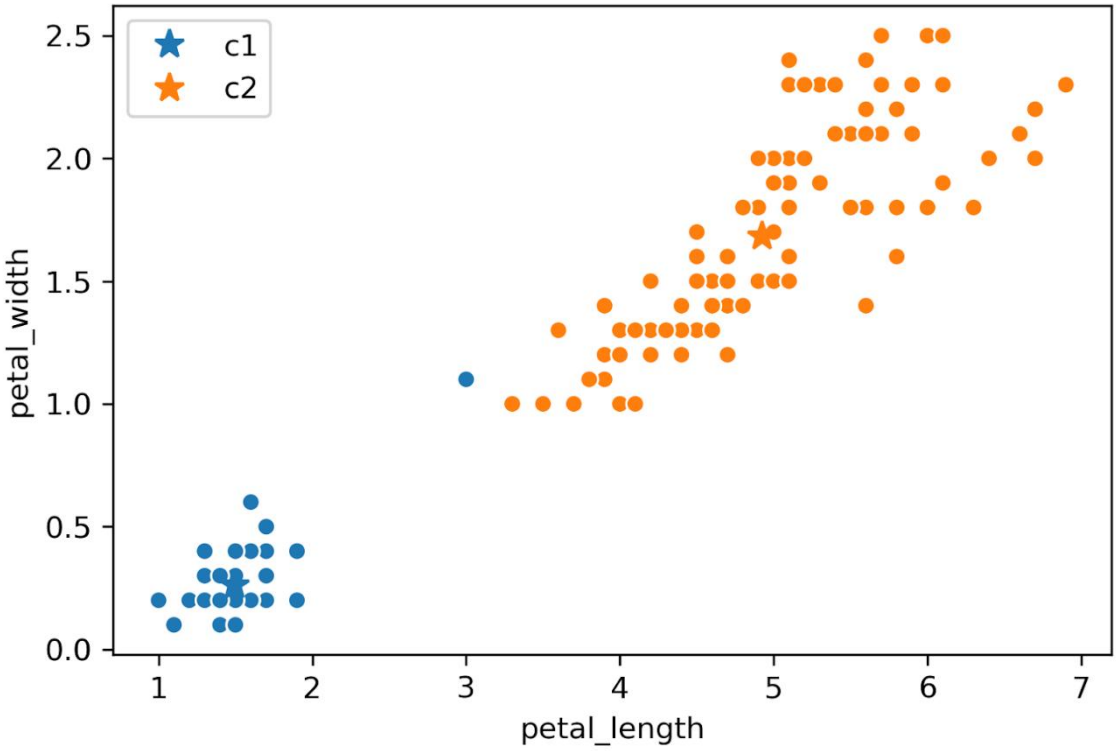
No better algorithm has been found ➔ K-Means is an NP-hard problem

Agglomerative Clustering

Which clustering result do you like better?





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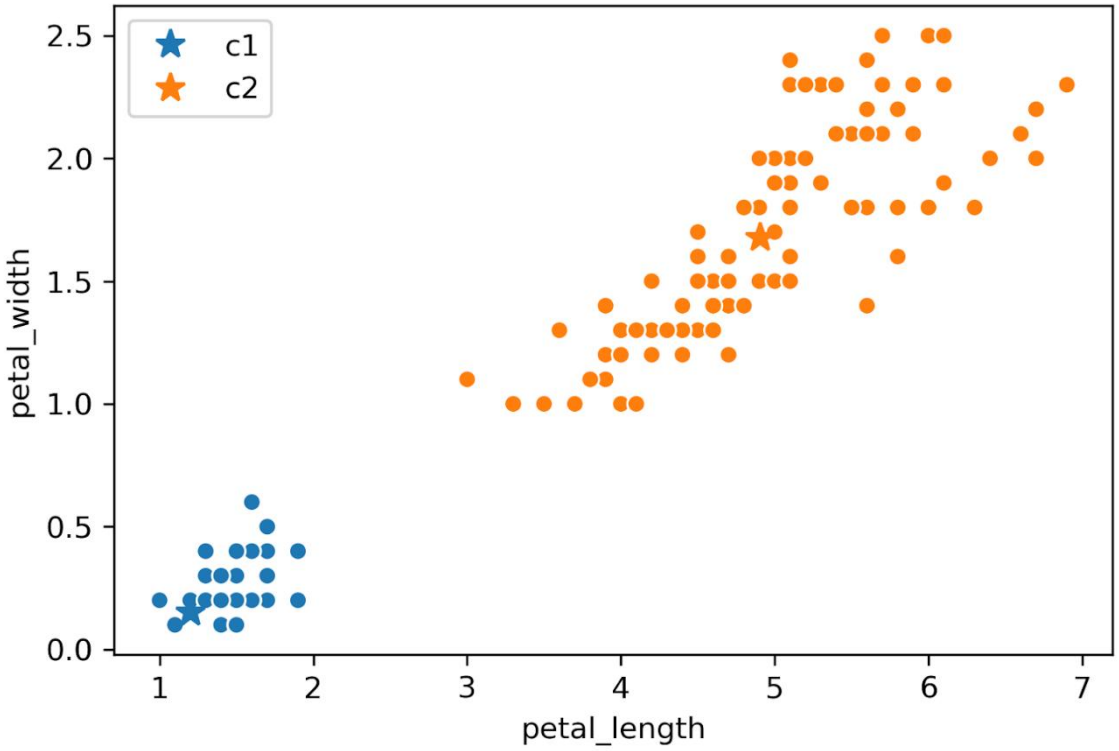
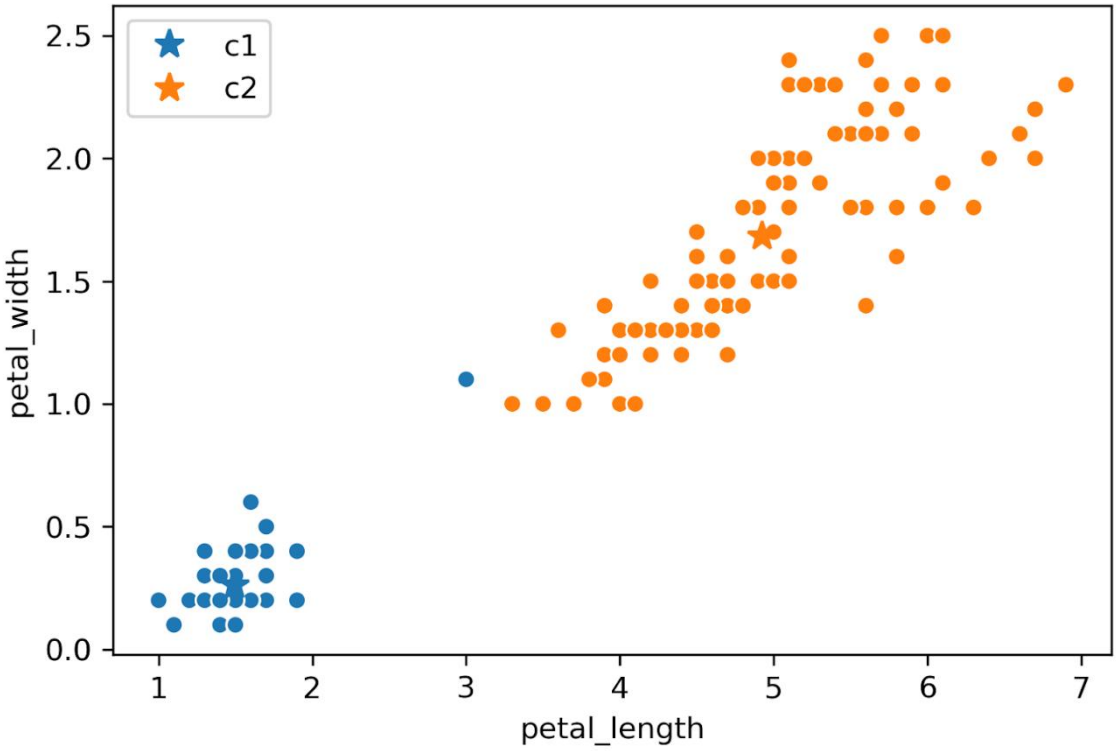




K-Means likes the one on the right better - it has lower inertia

- Why is the inertia on the right lower?
  - K-Means optimizes for distance, not “blobbiness”

Agglomerative Clustering

- As with regression and classification, there are many ways to do clustering

- Let's discuss an alternate idea known as "agglomerative clustering"

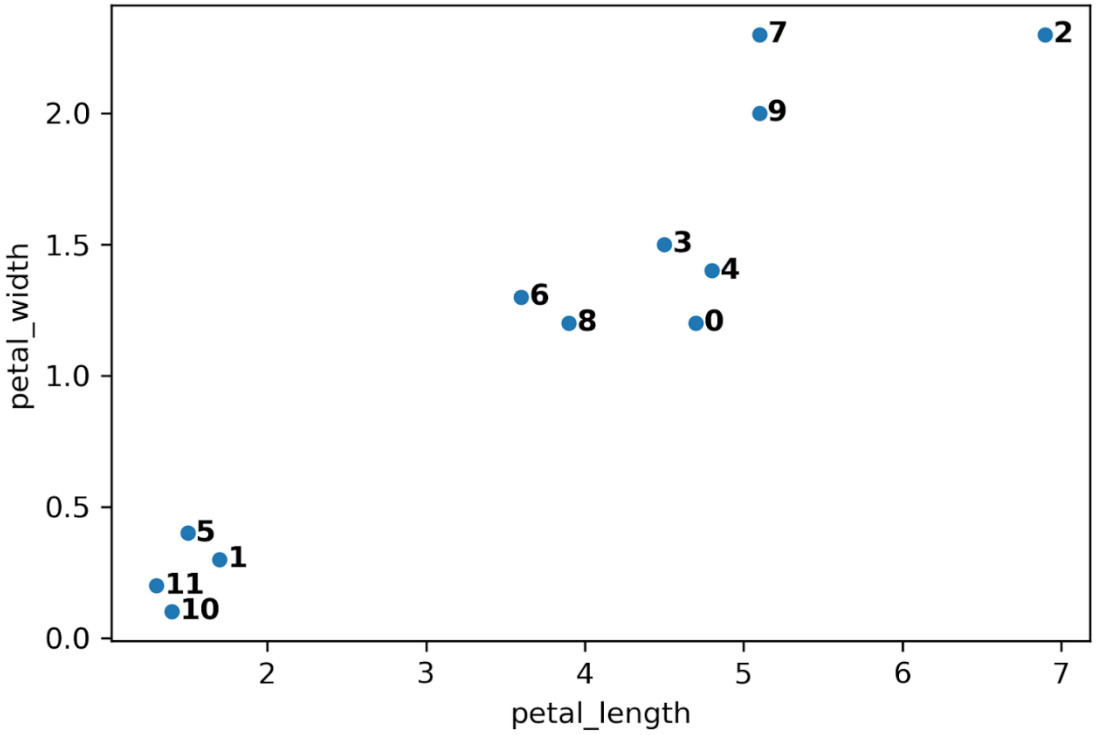
- Basic idea:
  - Every data point starts out as its own cluster
  - Join clusters with neighbors until we have only K clusters left

Let's see an example for K = 2

Agglomerative Clustering Example

When the algorithm starts, every data point is in its own cluster

- Below, 12 data points, so 12 clusters
- Closest clusters are 10 and 11, so merge them



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Two points in
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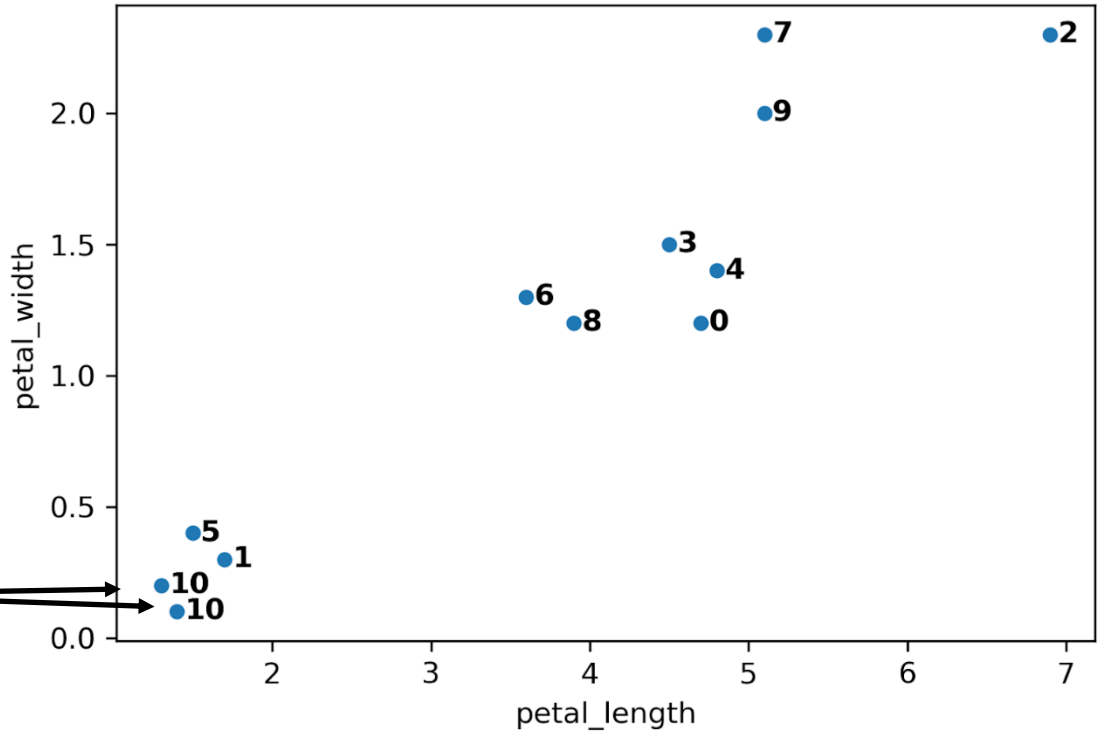

Agglomerative Clustering Example

The next two closest are 0 and 4, so merge them

Two points in
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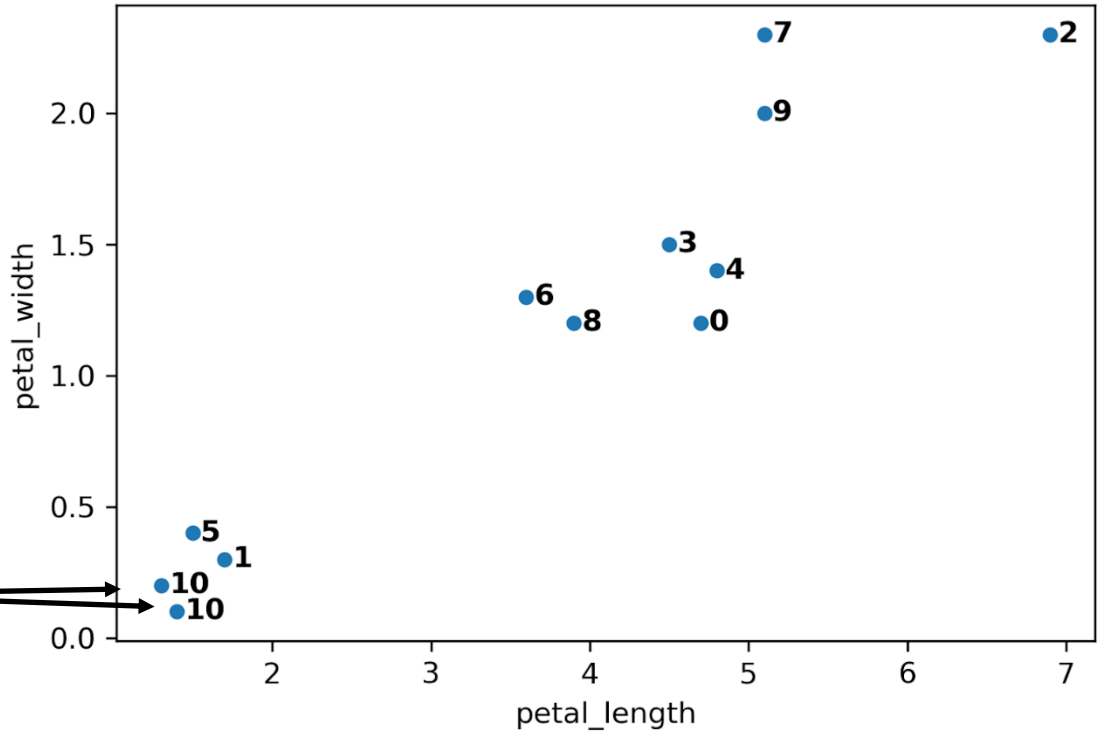

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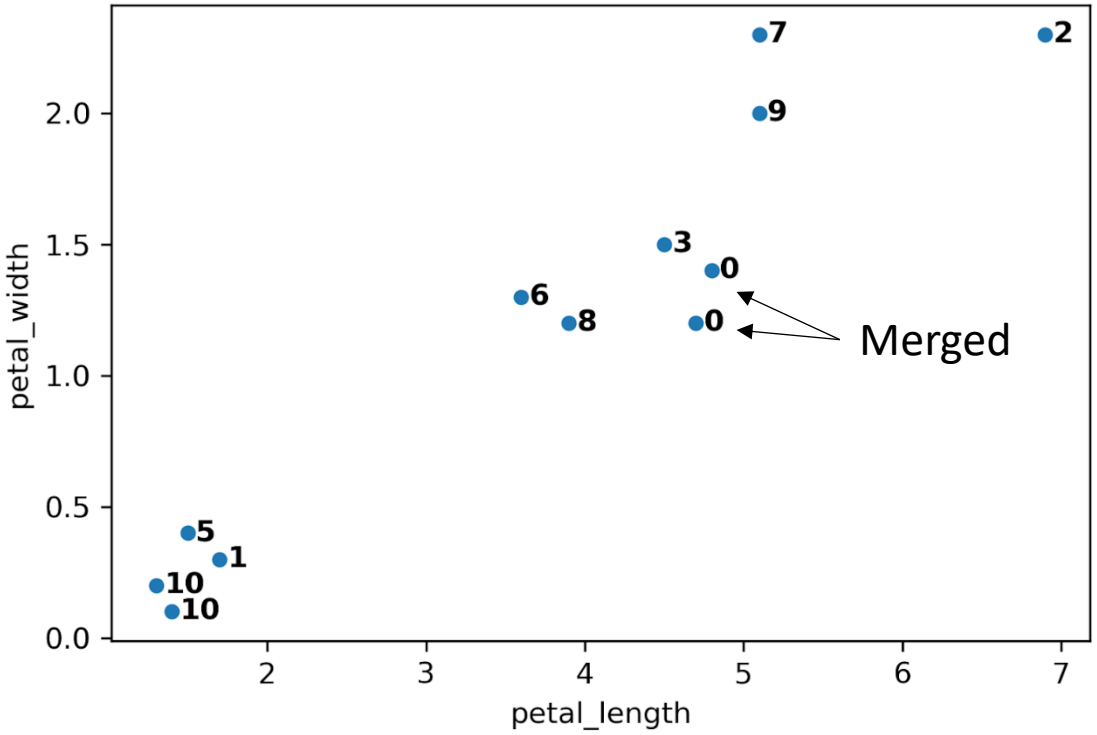
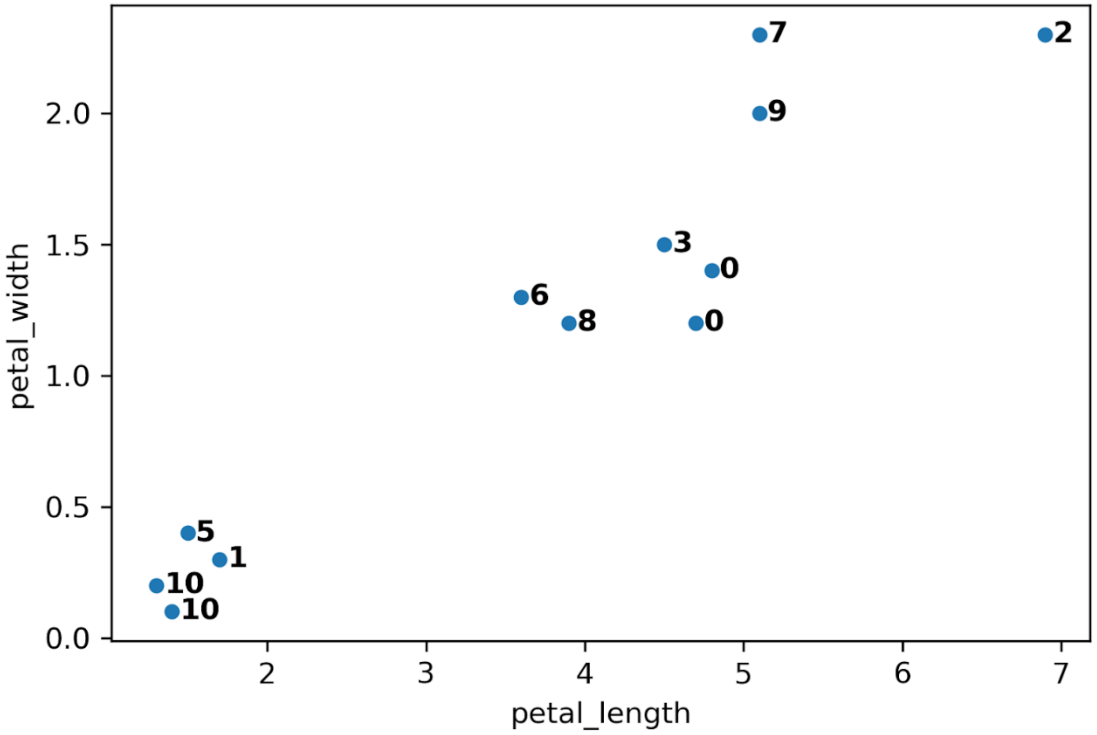


Agglomerative Clustering Example

At this point, we have 10 clusters:

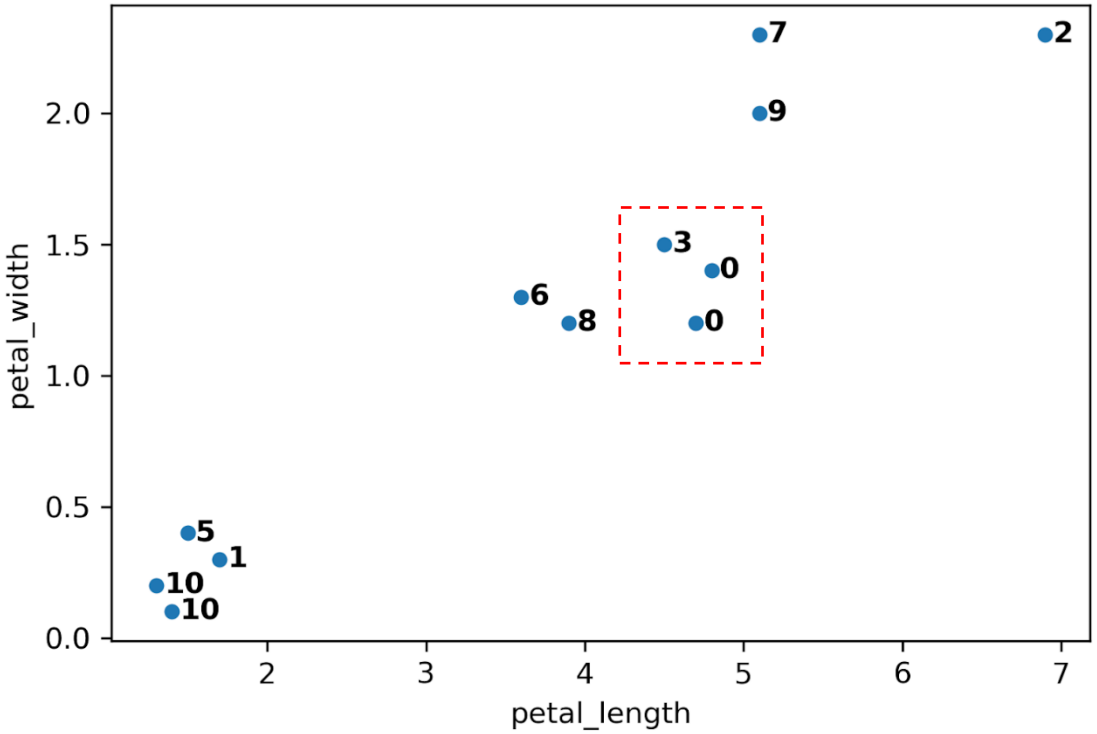
- 8 with a single point {1, 2, 3, 5, 6, 7, 8, 9}
- 2 with two points {0/0, 10/10}



Agglomerative Clustering Example

Tricky question:

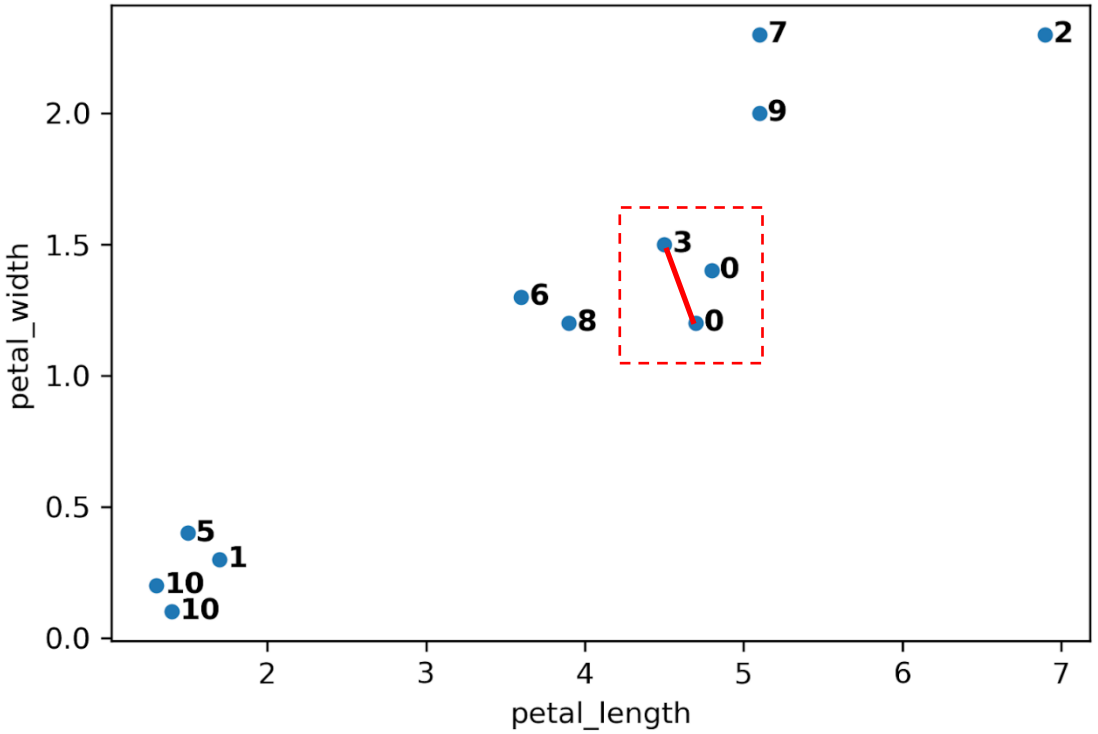
- What is the distance between clusters 0 and 3?



Agglomerative Clustering Example

Tricky question:

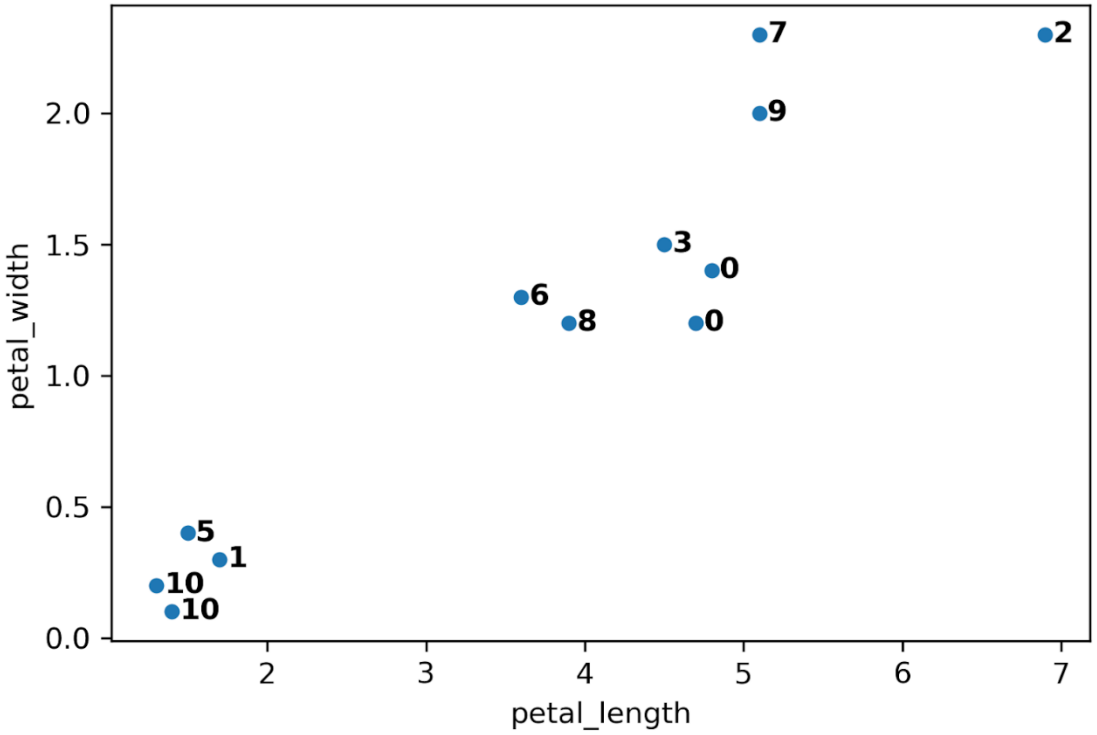
- What is the distance between clusters 0 and 3?
- There is no right answer - common choice is to use **max**



Would be perfectly reasonable to do something else - e.g., average or minimum

Agglomerative Clustering Example

The next two closest are 1 and 5, so merge them



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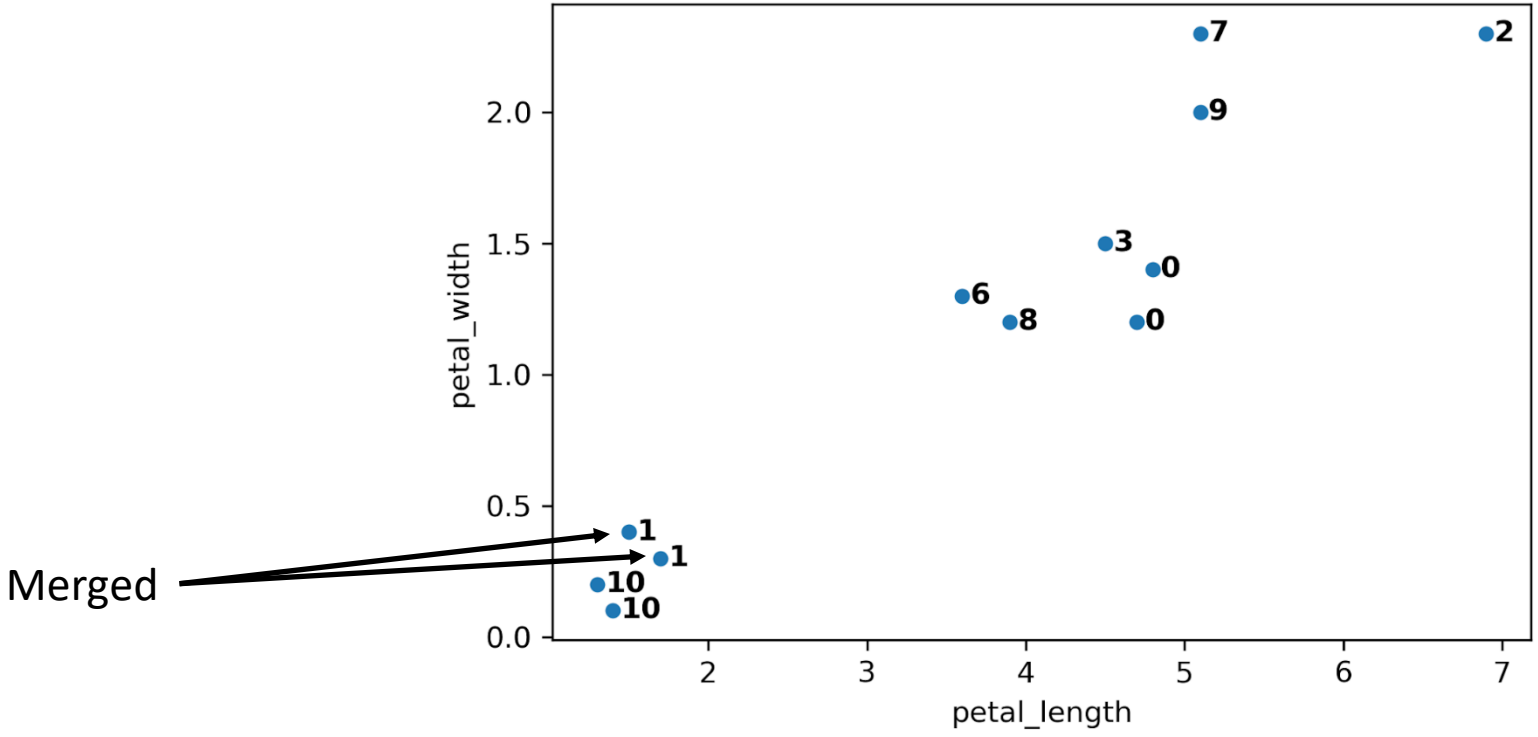
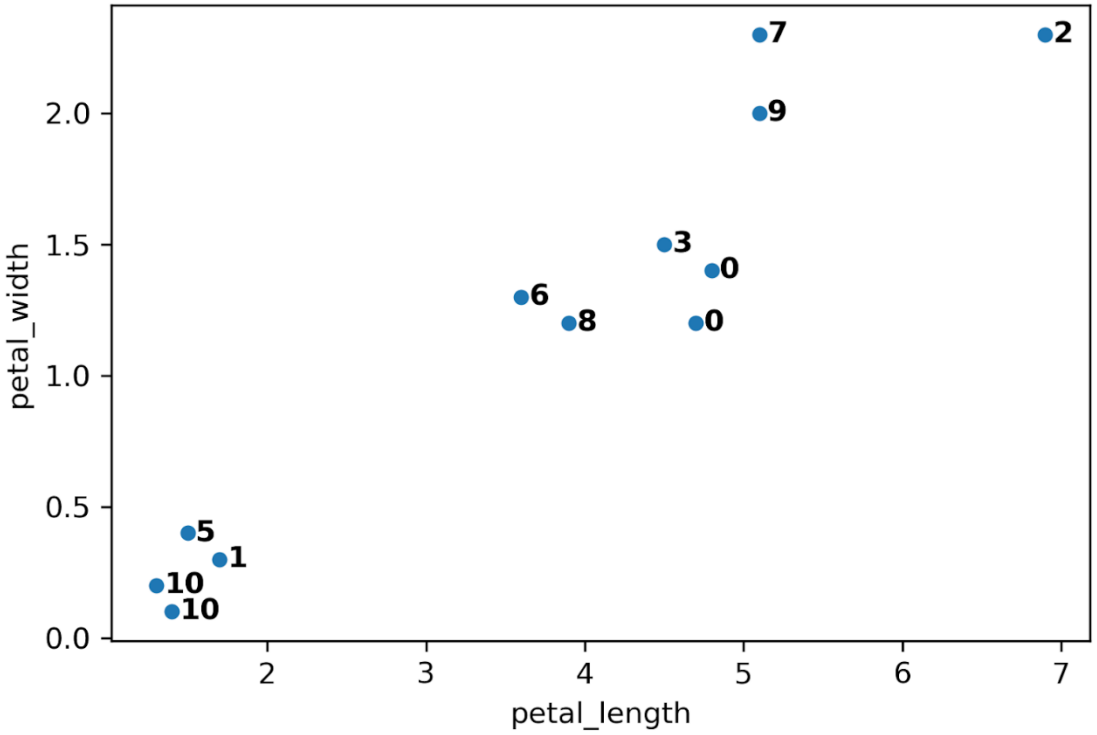


Agglomerative Clustering Example

The next two closest are 7 and 9

- It might not look that way, but the axes are not on the same scale!

The y-axis goes only up to 2.5, and the x-axis goes up to more than 7



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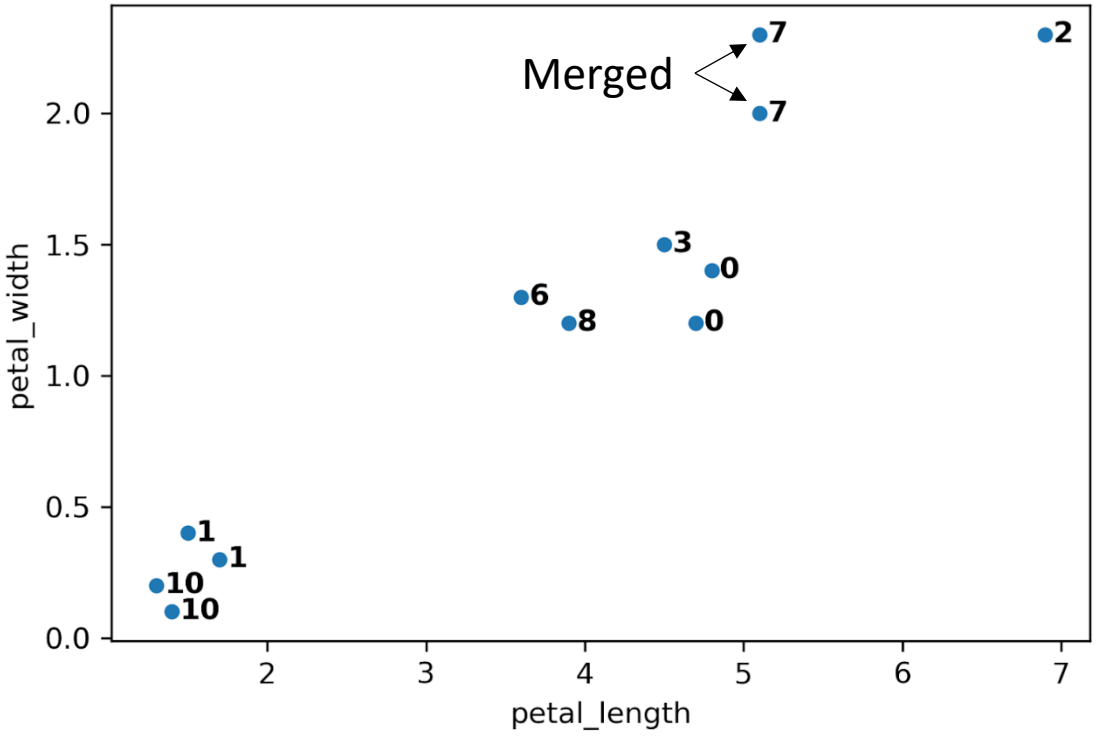

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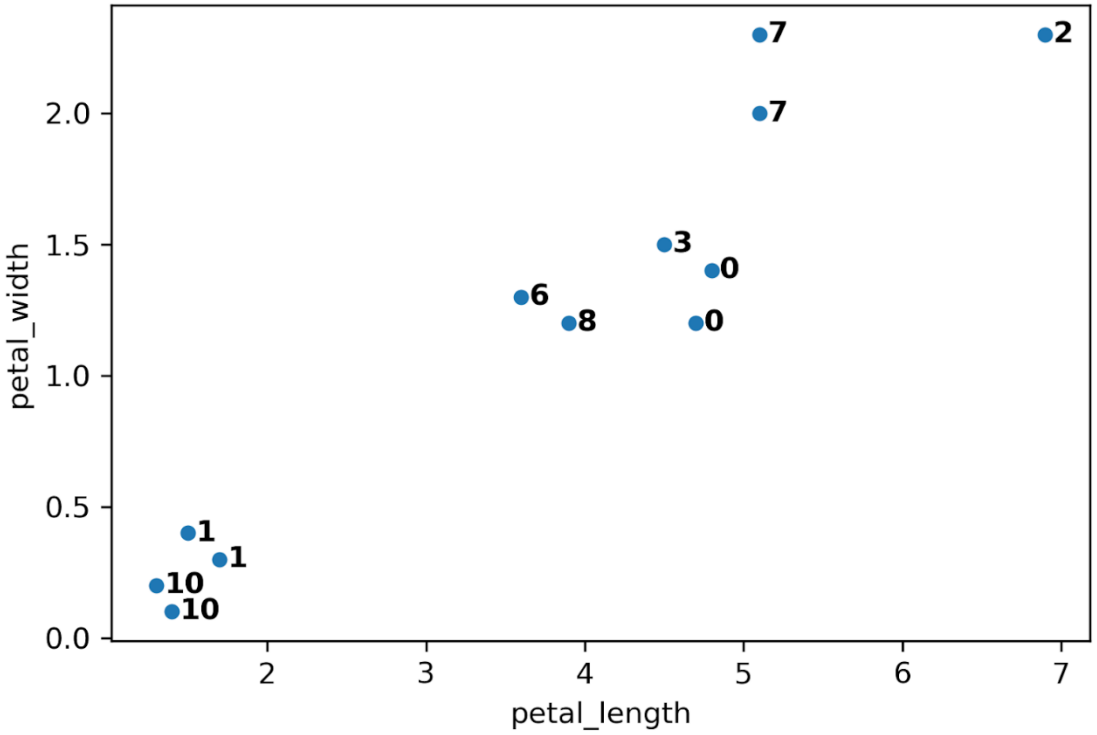

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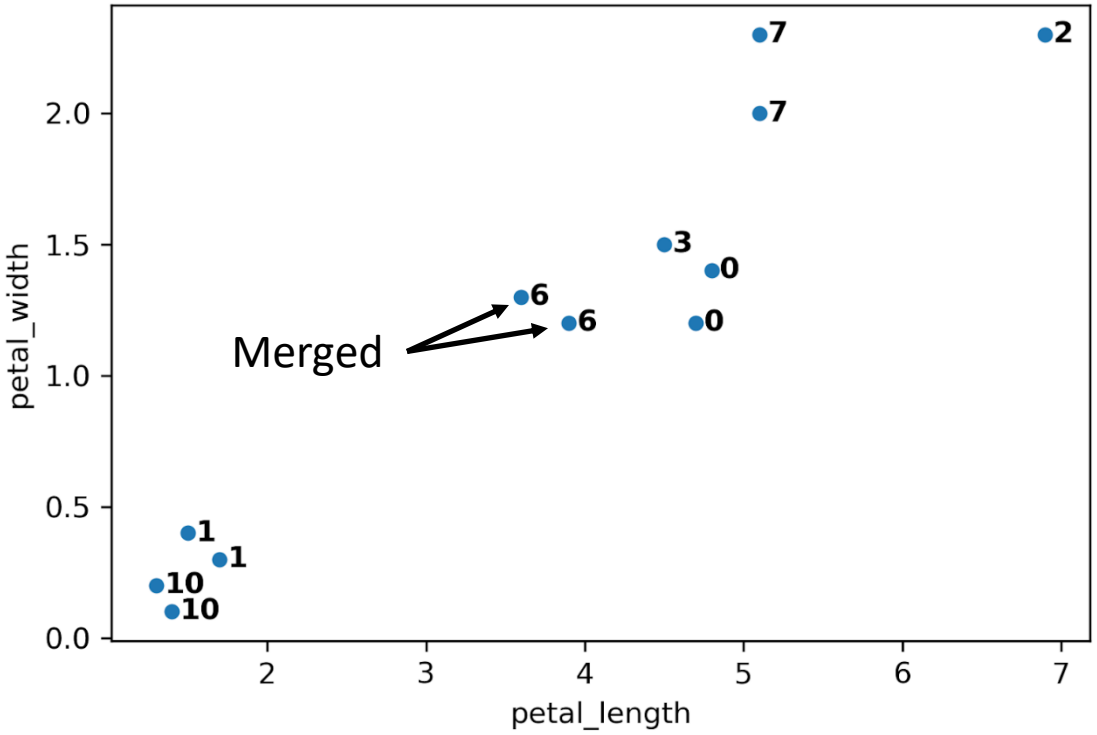
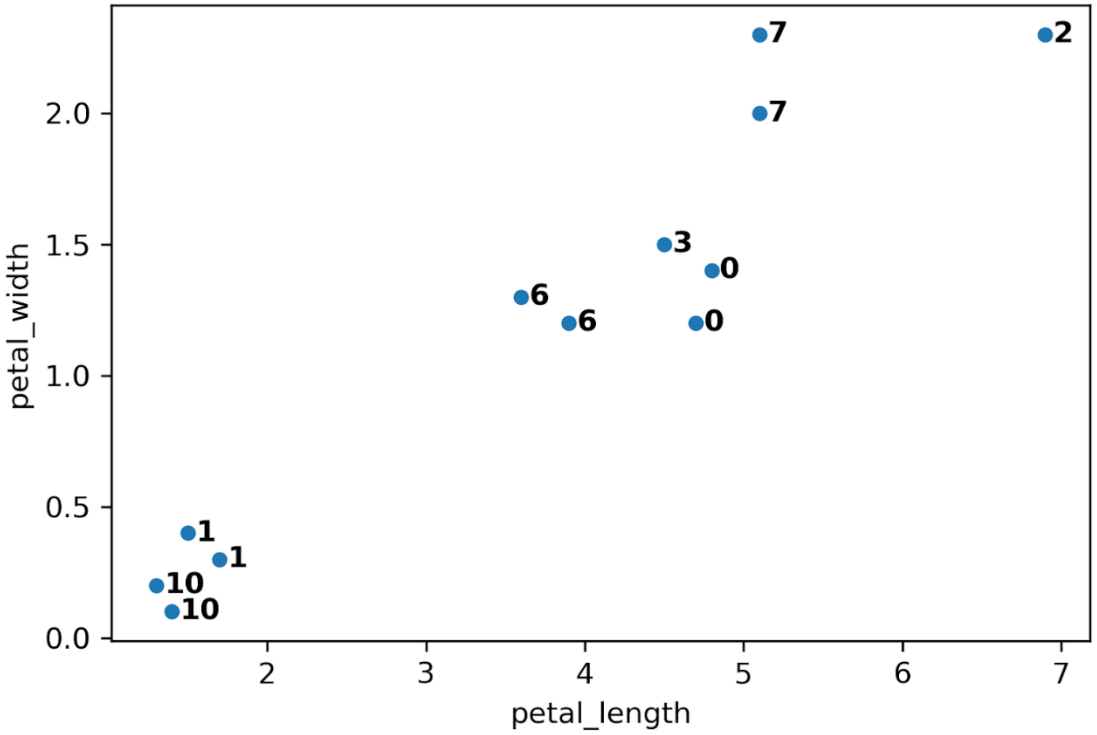


Agglomerative Clustering Example

The next two closest are 0 and 3



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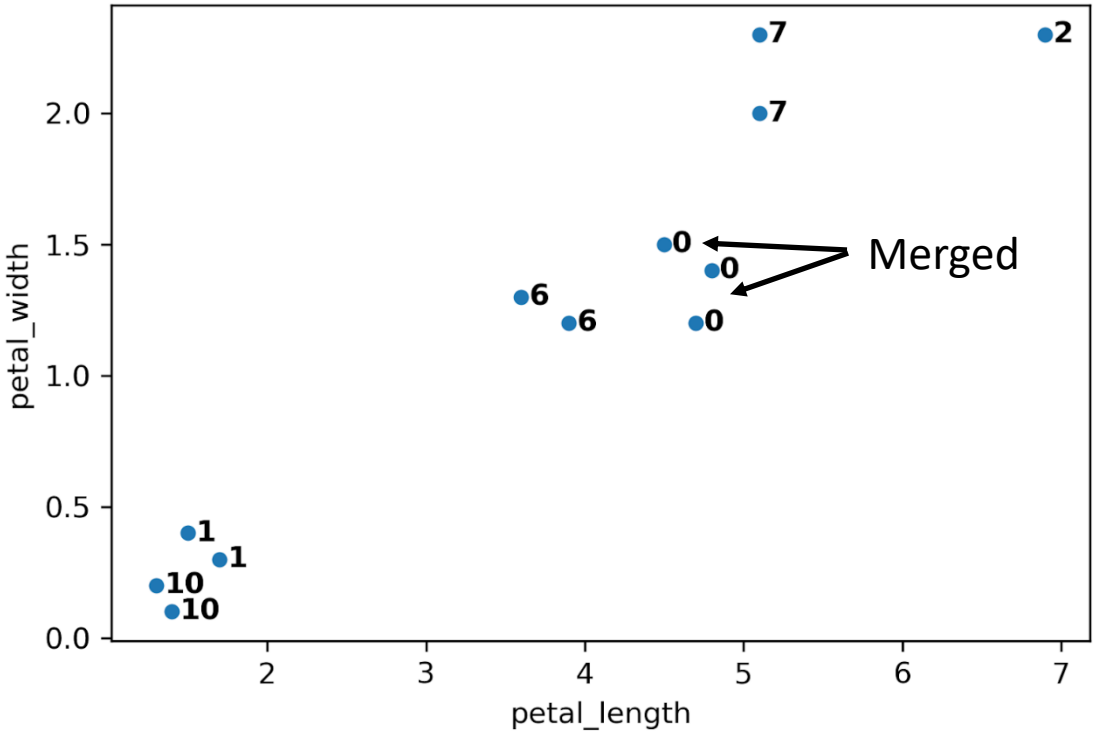
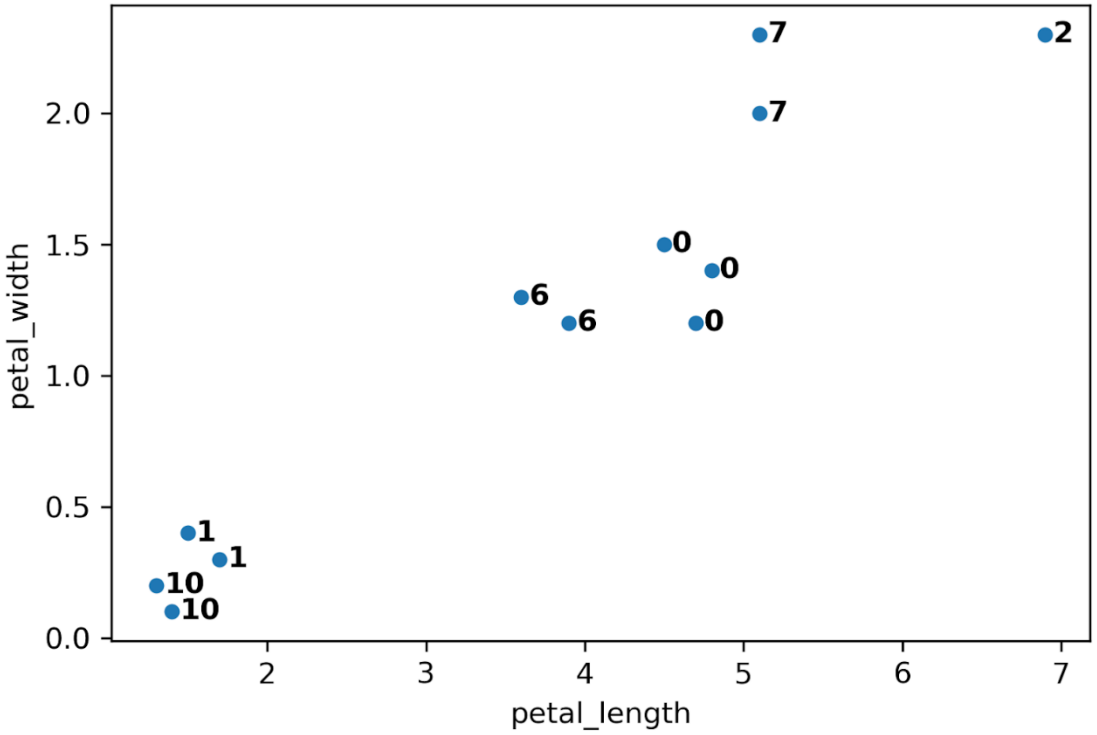


Agglomerative Clustering Example

The next two closest are 1 and 10



Agglomerative Clustering Example

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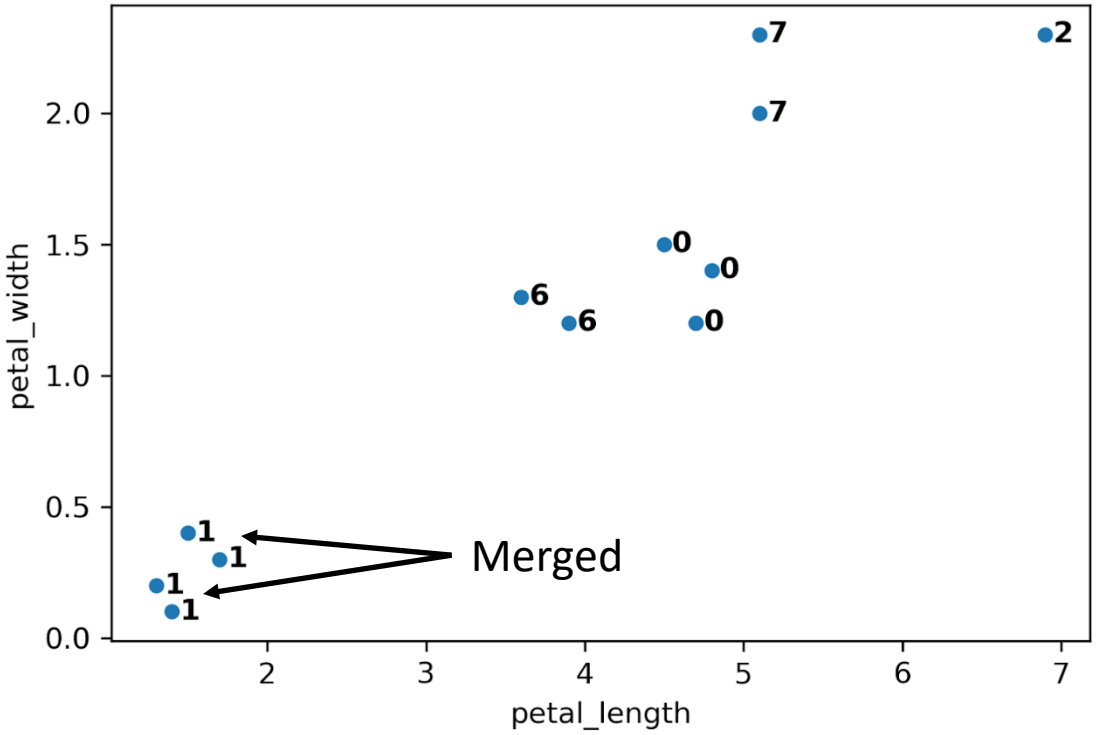
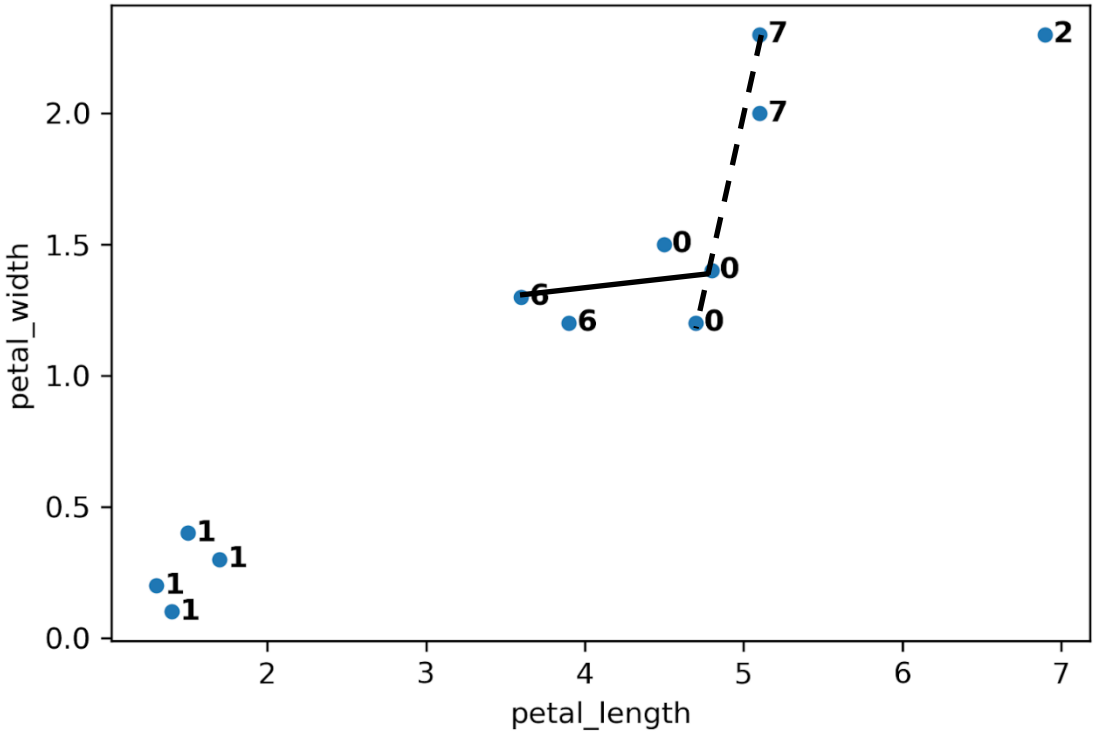


Agglomerative Clustering Example

The next two closest are 0 and 7.

- The max line between any member of 0 and 6 is longer than the max line between any member of 0 and 7



The dashed line is shorter than the solid line.

N.B. It doesn't look visually shorter due to different scales for x and y axes

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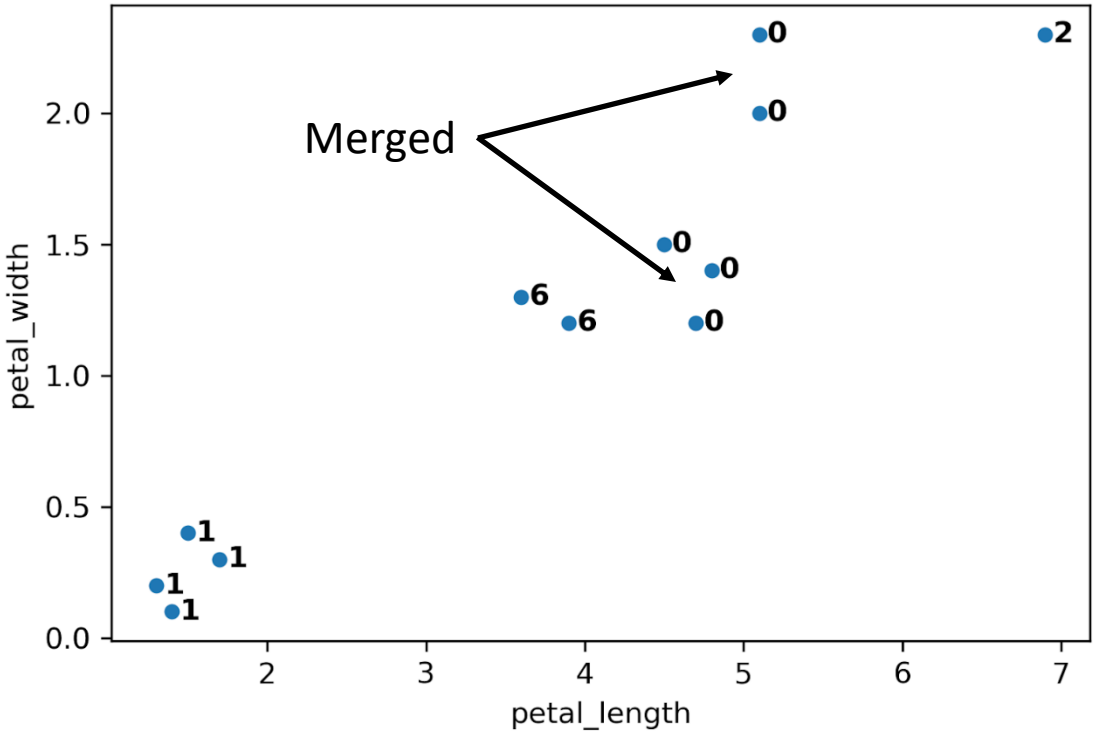

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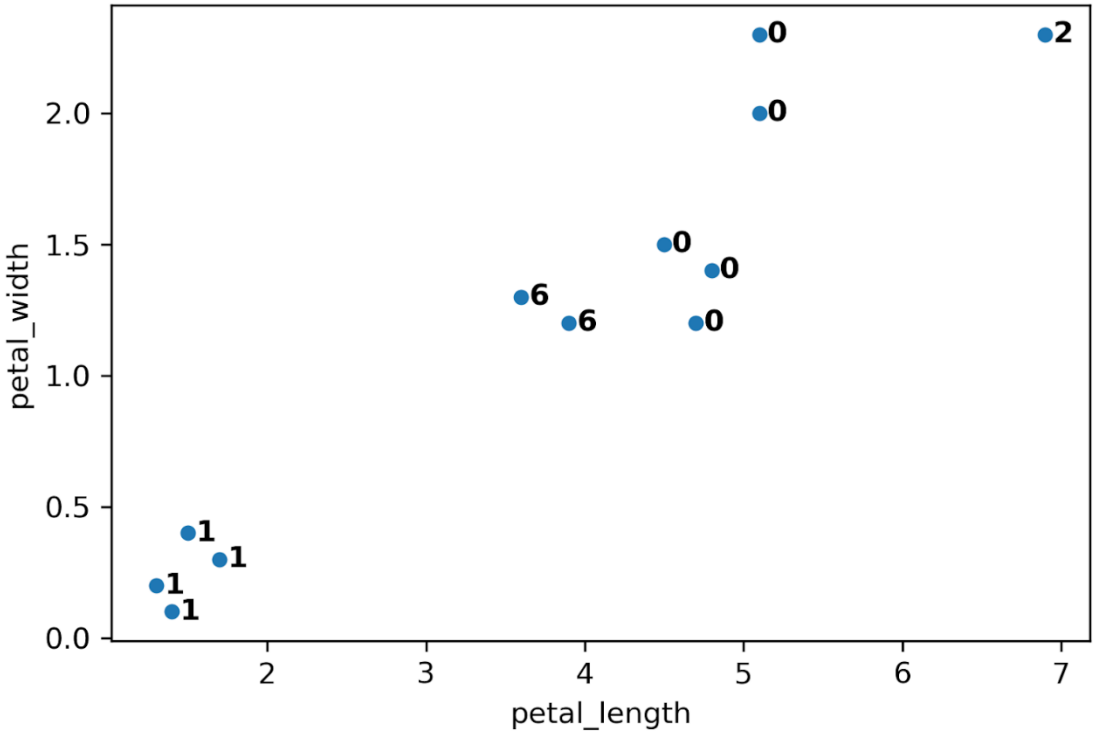

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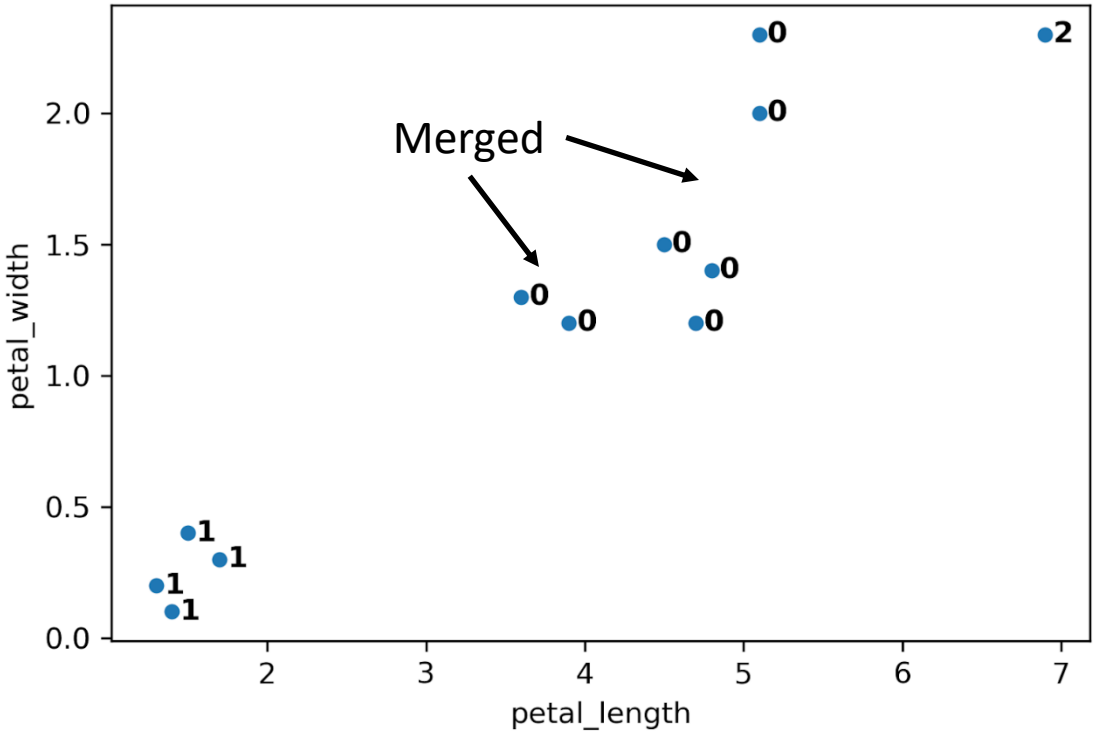
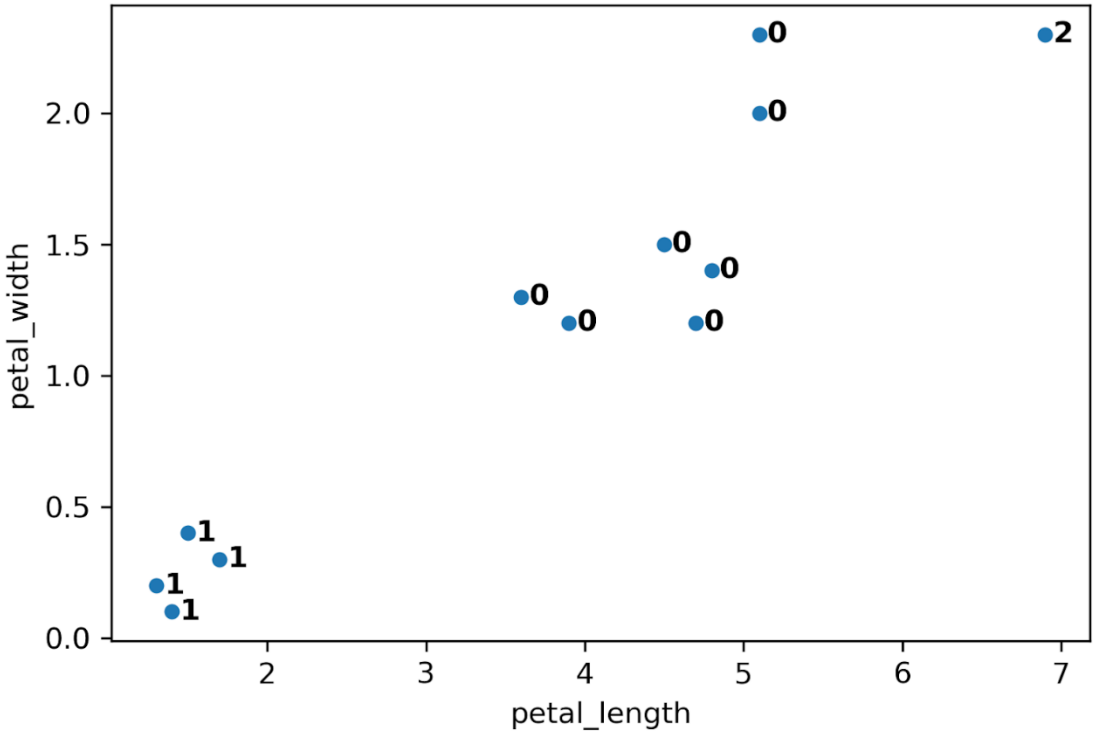


Agglomerative Clustering Example

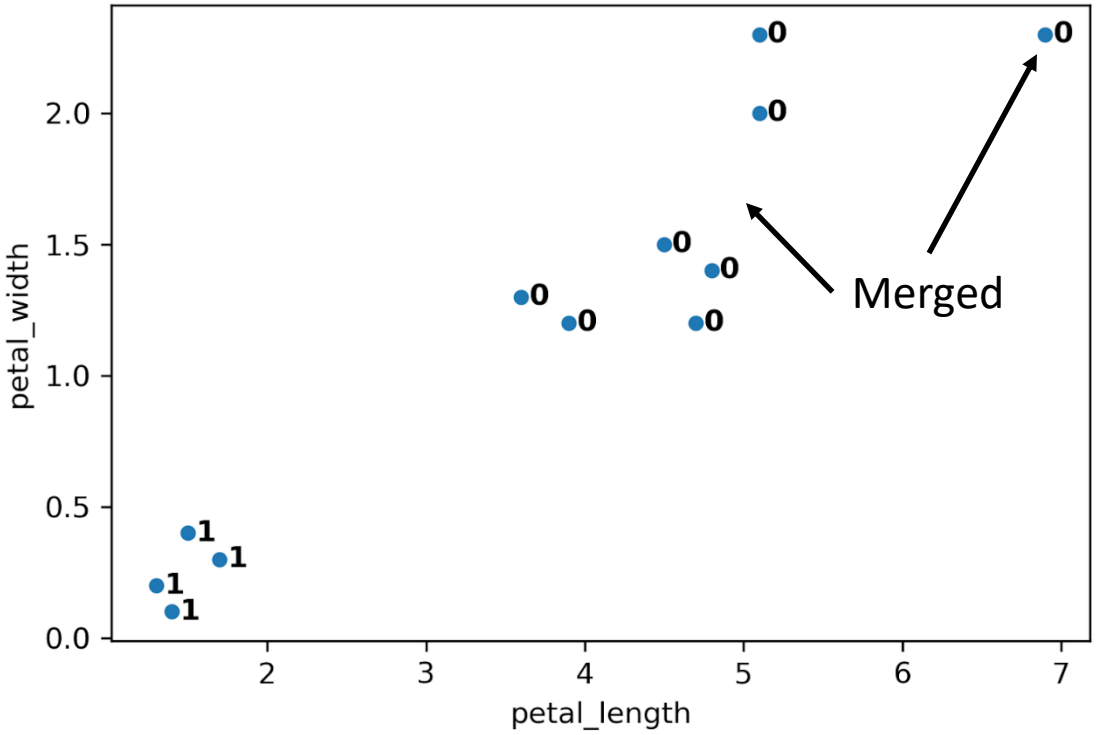
The next two closest are 0 and 2.



Agglomerative Clustering Example

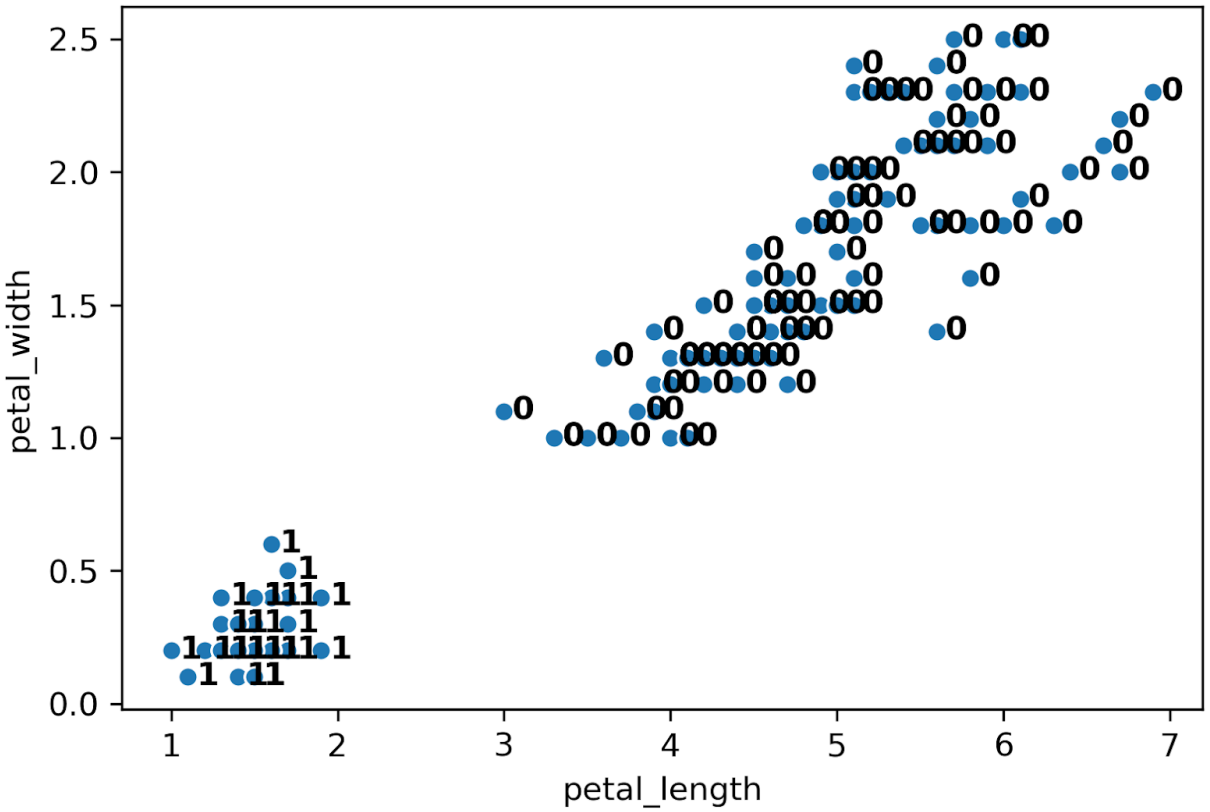
The next two closest are 0 and 2.

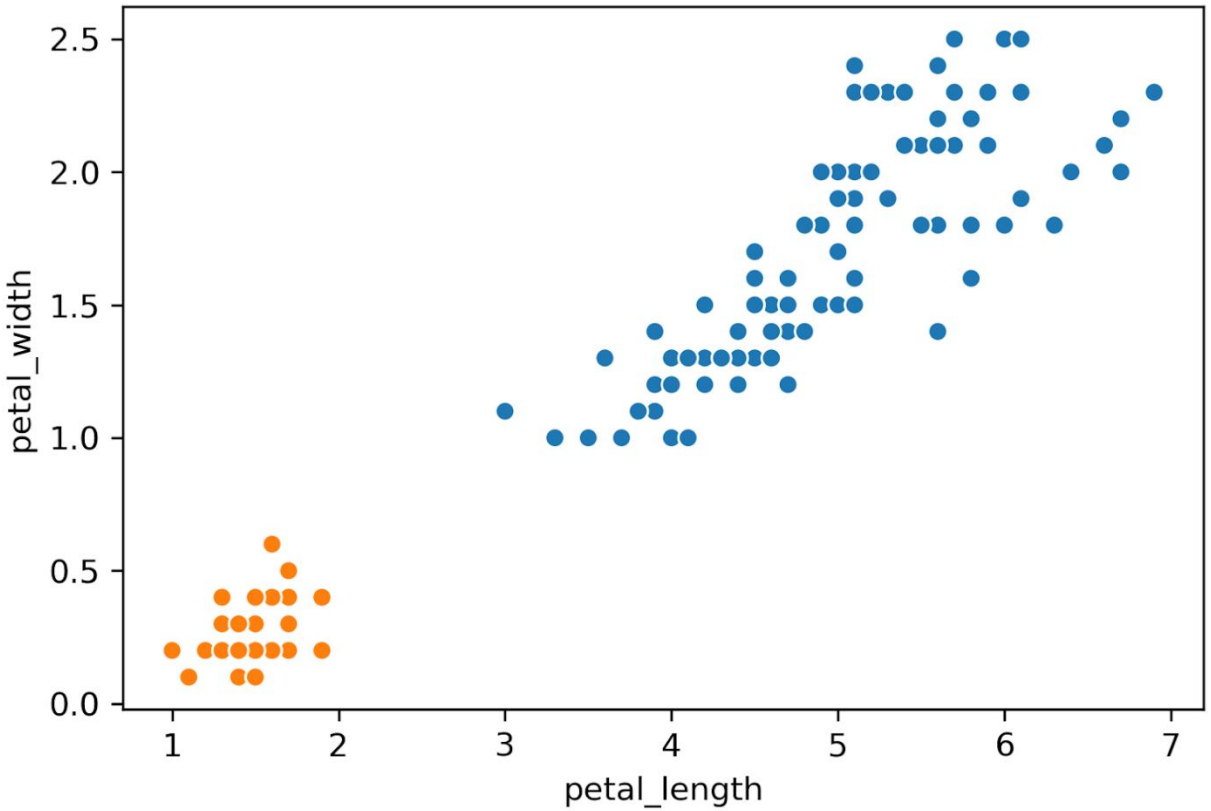
- At this point we are done, because we are left with only two clusters.



Agglomerative Clustering Example

On the full dataset, our agglomerative clustering algorithm gets the "right" output





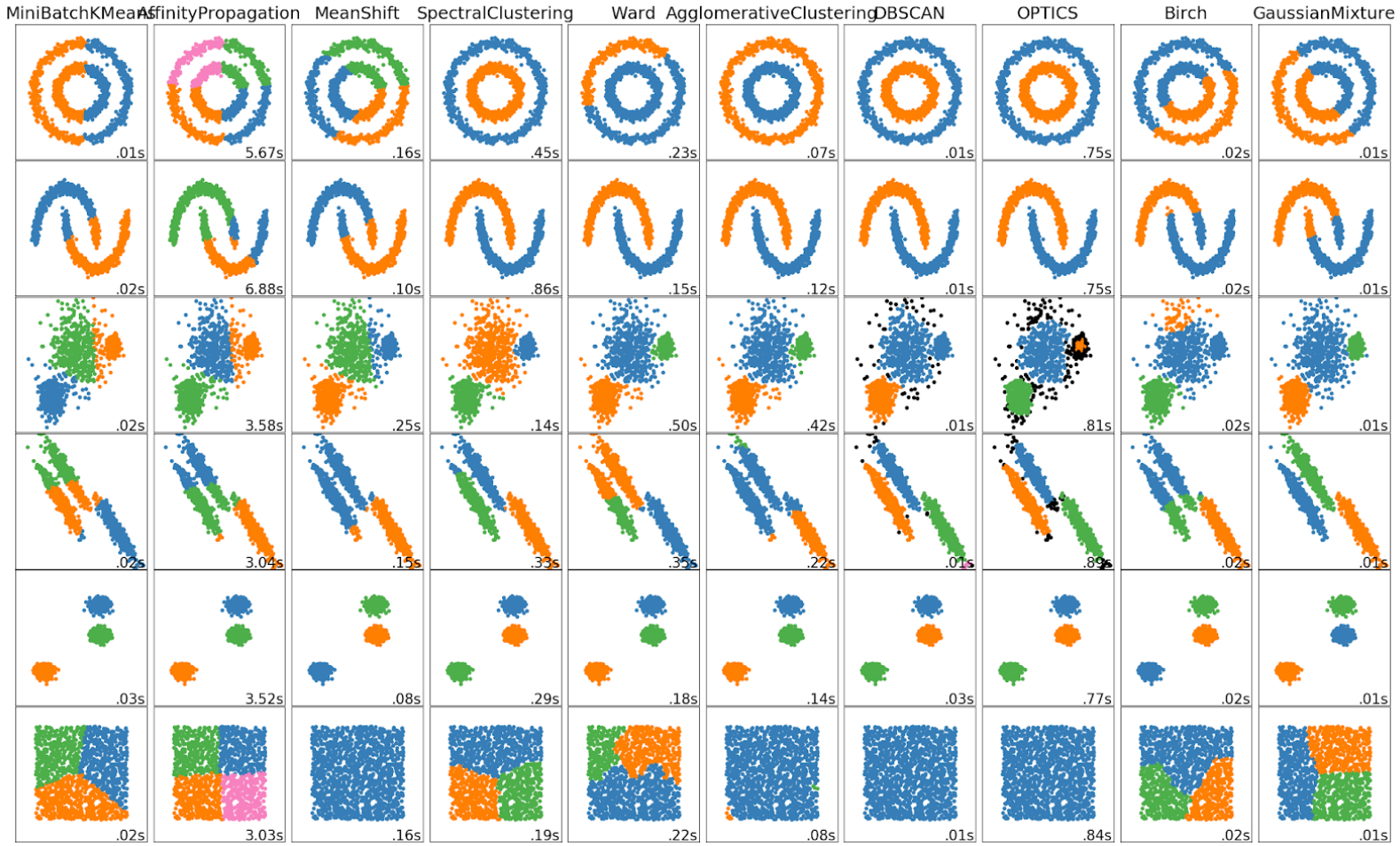
Clustering and scaling

Note that because clustering relies on distance to group observations, variables on **larger scales** will dominate the clustering algorithm.

Solution → Standardize variables before clustering

Notebook

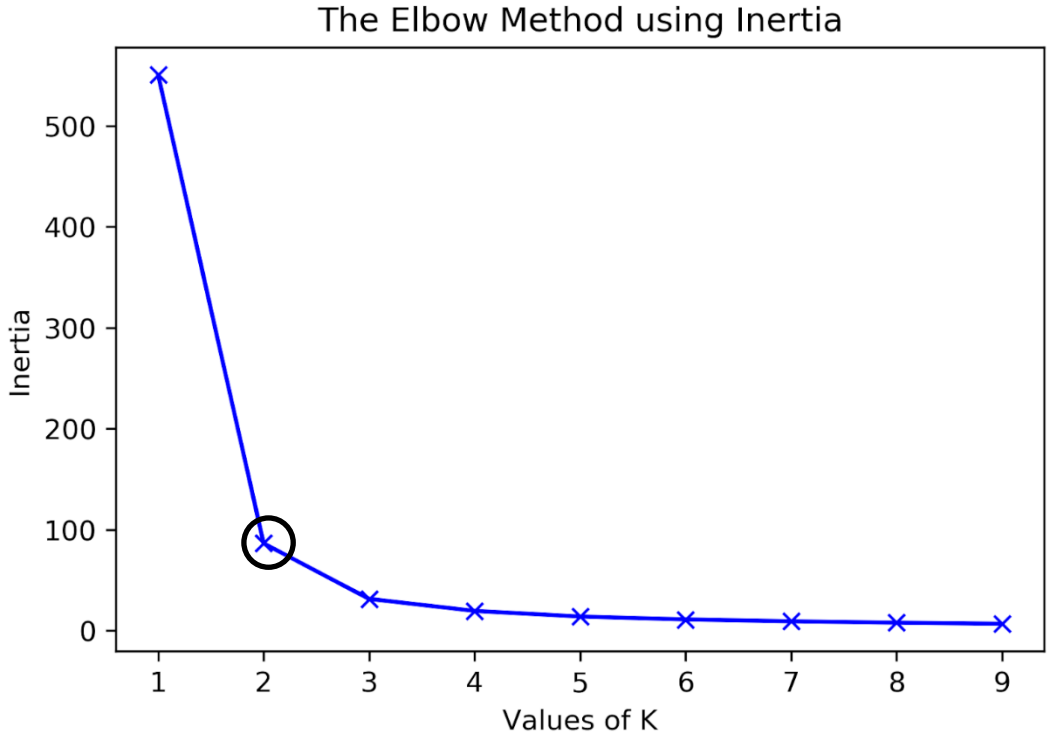
Clustering Algorithms: Many More We Haven’t Seen



https://scikit-learn.org/stable/_images/sphx_glr_plot_cluster_comparison_001.png

Picking K: the Elbow Method

- For K-Means, one approach is to plot inertia versus many different K values
- Pick the K in the “elbow”, where we get diminishing returns afterwards
- Note: Big complicated data often lacks an elbow



Picking K: Real World Metrics (Example from Andrew Ng)

Sometimes you can rely on real world metrics to guide your choice of K

Perform 2 clusterings:

- Cluster heights and weights of customers with K = 3 to design Small, Medium, and Large shirts
- Cluster heights and weights of customers with K = 5 to design XS, S, M, L, and XL shirts

To pick K:

- Consider projected costs and sales for the 2 different Ks
- Pick the one that maximizes profit

EDA and Clustering

Unsupervised learning in general is highly useful for EDA

- Naturally reveals structure and association in your data

Cluster multivariate data into some k-clusters, with no target in mind. Do distinct clusters emerge?

- If so, could mean there is latent categorical association in your data otherwise uncaptured by your measured variables.
- If not, then there might not be any such association.

EDA and Clustering

What could the following scenarios imply?

- Clustering human anatomical data reveals two distinct clusters

EDA and Clustering

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